



Daily high-resolution surface PM_{2.5} estimation over Europe by ML-based downscaling of the CAMS regional forecast

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ABSTRACT

Fine particulate matter (PM_{2.5}) is a key air quality indicator due to its adverse health impacts. Accurate PM_{2.5} assessment requires high-resolution (e.g., at least 1 km) daily data, yet current methods face challenges in balancing accuracy, coverage, and resolution. Chemical transport models such as those from the Copernicus Atmosphere Monitoring Service (CAMS) offer continuous data but their relatively coarse resolution can introduce uncertainties. Here we present a synergistic Machine Learning (ML)-based approach called S-MESH (Satellite and ML-based Estimation of Surface air quality at High resolution) for estimating daily surface PM_{2.5} over Europe at 1 km spatial resolution and demonstrate its performance for the years 2021 and 2022. The approach enhances and downscales the CAMS regional ensemble 24 h PM_{2.5} forecast by training a stacked XGBoost model against station observations, effectively integrating satellite-derived data and modeled meteorological variables. Overall, against station observations, S-MESH (mean absolute error (MAE) of 3.54 µg/m³) shows higher accuracy than the CAMS forecast (MAE of 4.18 µg/m³) and is approaching the accuracy of the CAMS regional interim reanalysis (MAE of 3.21 µg/m³), while exhibiting a significantly reduced mean bias (MB of -0.3 µg/m³ vs. -1.5 µg/m³ for the reanalysis). At the same time, S-MESH requires substantially less computational resources and processing time. At concentrations >20 µg/m³, S-MESH outperforms the reanalysis (MB of -7.3 µg/m³ and -10.3 µg/m³ respectively), and reliably captures high pollution events in both space and time. In the eastern study area, where the reanalysis often underestimates, S-MESH better captures high levels of PM_{2.5} mostly from residential heating. S-MESH effectively tracks day-to-day variability, with a temporal relative absolute error of 5% (reanalysis 10%). Exhibiting good performance at high pollution events coupled with its high spatial resolution and rapid estimation speed, S-MESH can be highly relevant for air quality assessments where both resolution and timeliness are critical.

1. Introduction

Fine particulate matter of size <2.5 µm (PM_{2.5}) has adverse effects on human health and the environment. Elevated levels of PM_{2.5} can contribute to the formation of haze, thus reducing visibility and disrupting daily life (Wu et al., 2020). Deposition of particulate matter into lakes can also negatively affect the water quality and clarity (Chen et al., 2014). Furthermore, PM_{2.5} has a severe impact on human health. Short-term exposure to high PM_{2.5} levels increases the likelihood of cardiopulmonary diseases (S. Yang et al., 2019), while long-term exposure is linked to premature deaths from cardiovascular, cerebrovascular, and respiratory diseases (Liang et al., 2018; Pope III et al., 2019). Despite the efforts to decrease emissions, around 97% of the urban population in the European Union (EU) continues to be exposed to PM_{2.5} concentrations above the World Health Organization (WHO) guidelines in 2022 (EEA, 2024). While fossil fuel combustion remains the main anthropogenic source of PM_{2.5} pollution, natural events such as

wildfires, dust storms, and stable meteorological conditions have been increasingly contributing towards elevated levels of PM_{2.5} (Cheng et al., 2016; Sokhi et al., 2022). Over Europe, agricultural emissions and residential heating also contribute substantially to high PM_{2.5} levels (EEA, 2022; McDuffie et al., 2020). An increase in PM_{2.5} levels in the future is also predicted due to the influence of climate change (Coelho et al., 2021). Due to these ongoing problems coupled with the availability of suitable observations and modelling data that can be used to support our study, we focus our study on the problem of PM_{2.5} pollution within Europe.

Due to its severe health consequences, PM_{2.5} assessment forms a part of the EU's legal framework on ambient air quality assessment and monitoring. European member states thereby officially report and implement air pollution control strategies to meet the air quality legal requirements (European Union, 2008; Miranda et al., 2015). The European Environmental Agency (EEA) and national authorities responsible for air quality management within Europe primarily utilize station

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measurements for annual assessments of air quality and accordingly design policies to reduce air pollution effects (EEA, 2020, 2021, 2022, 2023b, 2024).

A further component of the EU's air quality management strategy is the definition of a series of threshold concentrations for different air pollutants. The EU's air quality directive 2008/50/EC establishes both daily and annual PM_{2.5} thresholds to serve as indicators of air quality standards (European Union, 2008). In the newly revised (accepted spring 2024) directive, these thresholds are set at 25 µg/m³ for daily means and 10 µg/m³ for annual means, with the previous 2008 directive defining only an annual threshold of 25 µg/m³ (European Union, 2022). The air quality monitoring network within Europe and the use of the datasets it provides is therefore of critical importance for current air quality assessment and management. Despite the good density of PM_{2.5} monitoring networks over certain regions in Europe, their distribution is still inconsistent with regard to the pollution levels across Europe. Further, the monitoring stations themselves have limited spatial representation which can hinder the accurate mapping of spatio-temporal patterns and understanding of the actual impacts of PM_{2.5} pollution on human health (Manisalidis et al., 2020).

Due to the spatial limitation of air quality networks, other sources such as satellite observations of Aerosol Optical Depth (AOD) from instruments like Moderate Resolution Imaging Spectroradiometer (MODIS) or the Visible Infrared Imaging Radiometer Suite (VIIRS) have been widely used in the literatures to estimate surface PM_{2.5} concentrations using different techniques. These involve scaling approaches (Liu et al., 2004) and a range of statistical methods such as univariate regression (Wang and Christopher, 2003), multivariate regression (Gupta and Christopher, 2009), geographically weighted regression (Hu et al., 2013; Song et al., 2014; van Donkelaar et al., 2010; Van Donkelaar et al., 2016; H. Zhang and Kondragunta, 2021) and machine learning (ML) models (Hu et al., 2017; Ibrahim et al., 2022; Jiang et al., 2021; Y. Shao et al., 2020; Wei et al., 2020, 2021; Q. Xiao et al., 2018; N. Yang et al., 2022; T. Zhang et al., 2018). Additionally, very few studies have also implemented stacked ML approaches to integrate predictions from multiple models to obtain more accurate estimates compared to a single ML model (e.g., Chen et al., 2019). Using satellite AOD as the main source for estimating PM_{2.5} also has some disadvantages. Satellites provide a columnar AOD measurement and not that of the surface. Also, despite the good spatial coverage from satellites, the spatial continuity of satellite AOD is often negatively affected by clouds and highly reflective surfaces. This has led some studies into combining AOD information from satellites and Chemical Transport Models (CTMs) for a more continuous and accurate PM_{2.5} mapping (Geng et al., 2015; Goldberg et al., 2019).

CTMs, such as those used within the Copernicus Atmosphere Monitoring Services (CAMS), provide spatially continuous forecasts and concentrations of various atmospheric pollutants such as PM_{2.5} at regional and global scales. These products are also used to support assessment and monitoring of air quality. Since CTMs consider different chemical and physical mechanisms, they have a solid theoretical basis (Y. Zhang et al., 2021). However, the simulation process can be very complex, and expensive, and is typically at a relatively coarse spatial resolution of ~10 km. Additionally, the uncertainties of the input emission sources and meteorological conditions also impact their accuracies (H. Zhang et al., 2022). Some recent studies have implemented different techniques to downscale and improve the accuracy of CTM forecasts without the need for data assimilation. Particularly, ML methods have been increasingly used in surface PM_{2.5} mapping and forecast bias corrections due to their ability to map complex and non-linear relationships. ML approaches such as eXtreme Gradient Boost (XGBoost) have proven to be computationally fast and require only modest computational infrastructure (Tang et al., 2024). Since ML methods are usually trained on historical datasets and do not require station observations at the prediction stage, they offer the advantage of increased timeliness by not relying on real-time data from monitoring

stations. Studies such as Bi et al. (2022) used ML methods like Random Forest (RF) to generate 5-day forecasts of PM_{2.5} at 1 km resolution from NASA's Goddard Earth Observing System "Composition Forecasting" (GEOS-CF) with improved accuracy over China. Similarly, Salah Eddine et al. (2024) evaluated the performance of various ML models such as RF, XGBoost, Support Vector Regression, and Generalized Additive Models in improving the accuracy of CAMS PM₁₀ forecast over Morocco. Varga-Balogh et al. (2020) followed a multi-model fusion method to downscale the CAMS PM_{2.5} forecast over monitoring sites in Budapest. Similarly, Bertrand et al. (2023) explored the use of ML methods to reduce the systematic bias observed in CAMS ensemble models over monitoring stations. Some studies such as Dhandapani et al. (2023) and Gupta et al. (2021) have used the reanalysis aerosol products from Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA2) to estimate surface PM_{2.5} using ML over India and Thailand respectively. Few other studies have adapted the more complex deep learning methods for surface PM_{2.5} estimation using CTMs. For instance, Geiss et al. (2022) explored the potential of deep convolution neural networks for downscaling atmospheric simulations through its application on NASA's GEOS-CF and identified higher accuracy of this approach over the conventional downscaling techniques. Kaffashzadeh & Aliakbari Bidokhti (2024) used a neural network to improve the temporal understanding of ozone variability over monitoring stations in Iran by downscaling ozone product from CAMS global forecast and reanalysis, with the methodology that can be extended to other chemical species such as aerosols. Y. Xiao et al. (2022) developed a deep-learning framework called DeepCAMS for the spatio-temporal downscaling of GEOS-CF PM_{2.5} product. Most of these studies have focused on improving the temporal forecast accuracy or the model outputs at specific point locations or downscaling reanalysis products.

CAMS products are also used to support the assessment and monitoring of air quality within Europe. For example, the CAMS regional interim reanalysis (a data product mapping European air quality based on the data assimilation of surface observations within the CAMS forecasts) is used to evaluate air quality on an annual basis (Hamer et al., 2023; Tarrasón et al., 2022; CAMS, 2024c). In these cases, the annual and daily mean concentration indicators for PM_{2.5} are used to provide support for national reporting of the air quality situation in the previous calendar year. Two weaknesses of the CAMS interim reanalysis in this regard are their timeliness (only available on an annual basis each March for the preceding calendar year) and spatial resolution (0.1° × 0.1°). For such air quality assessments of PM_{2.5}, a higher spatial resolution (at least 1 km) and increased timeliness are desirable. Increased spatial resolution can improve the relevance of the information for evaluating PM_{2.5} levels over urban areas and increased timeliness can deliver this important information to users on a faster basis to support air quality assessment and management. In this study, we address this gap in air quality assessment by downscaling the daily available CAMS regional PM_{2.5} forecast product to derive a downscaled PM_{2.5} hindcast product. Furthermore, to the best of our knowledge, no previous studies have explored the potential of ML and CTM forecasts for deriving a higher resolution daily hindcast PM_{2.5} products.

We present a new variant of our approach entitled S-MESH (Satellite and ML-based Estimation of Surface air quality at High resolution) exploiting the potential of a stacked XGBoost ML model to estimate daily averaged surface PM_{2.5} at 1 km spatial resolution over Europe for daily and annual indicators. The approach meets the need for both daily and annual PM_{2.5} indicators that are important due to their relevance in monitoring PM_{2.5} within the EU's regulatory framework. A variant of the S-MESH approach has been previously used for estimating satellite-based surface NO₂ (Shetty et al., 2024) and is here extended to PM_{2.5}. The current approach downscales and improves the CAMS regional PM_{2.5} forecast using satellite AOD from the VIIRS instrument, meteorology data, and other predictor variables as well as station observations (for training only), and its performance is demonstrated for the years 2021–2022. Unlike traditional satellite-based PM_{2.5} estimation methods

that primarily rely on columnar AOD data, the current approach also integrates CAMS forecast $PM_{2.5}$ as a key input in the stacked models. This integration not only allows us to account for physicochemical mechanisms but also helps address issues related to gaps in satellite AOD data. It should be noted here that the objective of our approach is not to produce a downscaled and bias-corrected forecast product, but to obtain a higher-resolution, bias-corrected daily averaged hindcast dataset that is comparable to the CAMS regional interim reanalysis $PM_{2.5}$ product in terms of its potential applications that can be made available in a much more timely manner, e.g., within weeks rather than on an annual basis. Existing methods focus largely on downscaling forecasts or reanalysis products in general, which do not address the need for timely analysis of pollution events. Thus, the output of the S-MESH approach complements CAMS products by supporting air quality assessment through fast and efficient production, improved timeliness, and higher spatial resolution.

2. Materials and methodology

The current study focuses on land regions across Europe within 26° W– 45° E and 34° N– 72° N, including countries that report air quality data to the EEA. The study period spans the years 2021 and 2022. To focus on the most recent years and based on the availability of certain datasets (CAMS regional reanalysis), these two years are selected for the study. The presented approach utilizes various datasets from multiple sources for training and evaluation. They are described in the following sections.

2.1. Datasets

Various factors such as meteorology, population density, land cover type, and satellite AOD can have an influence on and thus help to determine the $PM_{2.5}$ concentration levels (Tang et al., 2024; X. Xu et al., 2021). After exploratory analysis, based on simplicity and accuracy the below-described variables were best suited and considered in this study for training and evaluation of the model.

2.1.1. $PM_{2.5}$ station measurements

Surface $PM_{2.5}$ observations measured by ground-based stations distributed across the study area are reported by cooperating countries to the EEA. Around 36% of these stations are considered urban background sites and around 22% are urban traffic sites. The remaining 42% are primarily a mix of suburban, rural, and industrial sites. As a result of policy requirements from the EU Air Quality Directive (European Union, 2008), the density of these stations is closely linked to population density and thus relatively high in central Europe and comparatively low over, for example, Scandinavia. The stations measure ambient $PM_{2.5}$ concentration using different techniques (for example gravimetric reference instruments or different reference-equivalent instruments) to provide hourly measurements. The data is publicly available from the EEA e-Reporting database (EEA, 2023a). For our study, we acquired all available $PM_{2.5}$ surface observations for the years 2021–2022 from this database. To ensure quality-controlled data, we filtered the data to only include verified station observations which are indicated by the ‘Verification’ quality flag for each observation. These were subsequently aggregated into daily averages for this study. Based on the suggested criteria in the EU directive, those daily station measurements with at least 18 h of observation (>75% daily coverage) are included when aggregating for daily averages (European Union, 2008). A final set of 1,773 stations was used in this study and served as a training dataset for the ML model.

2.1.2. CAMS regional forecast and interim reanalysis

CAMS provides information about the atmospheric composition at global and regional scales through CAMS global and CAMS regional products (Marécal et al., 2015; Peuch et al., 2022). CAMS global products are at a coarser spatial resolution of ~ 40 km and are focused over

the global domain while CAMS regional products are at a comparatively higher spatial resolution of ~ 10 km and focused over the European domain. Due to their higher spatial resolution and Europe being our region of interest, the study uses CAMS regional products.

The CAMS regional production system provides a daily ensemble 4-day forecast, analysis, and reanalysis of main air pollutants such as NO_2 or $PM_{2.5}$ for the European domain from 11 individual state-of-the-art European regional atmospheric chemistry models (ECMWF, 2024). These individual models differ in their way of integrating meteorology and chemistry, usage of the number of chemical species, physical and chemical parameterization schemes, biogenic emissions, data assimilation methods, and have different native grid resolutions. These differences contribute to a substantial spread within the ensemble model. Hence the median of all individual models is provided as an ensemble product, which on average provides better performance than the individual models (Casciari et al., 2022). These regional air quality models use the European Centre for Medium-Range Weather Forecasts’s (ECMWF) Integrated Forecasting System (IFS) for the meteorological parameters, IFS global production for boundary conditions of chemical species, and custom regional emissions produced for CAMS (Kuenen et al., 2022).

The forecasts are combined with measurements from monitoring stations from the previous day to provide the analysis product for different pollutants. The assimilated station observations are near-real-time and as such not quality-controlled. Hence a revised analysis product of previous years is generated with the ensemble of different models but using more and quality-controlled station measurements and satellite observations along with a reanalysis of meteorological modeling. Carried out at the European scale, the result of this process is the CAMS regional reanalysis, which is available in two streams: An interim reanalysis product, which assimilates up-to-date (near real-time) station observations, is available with a delay of just over one year. A validated reanalysis product, which assimilates validated station observations, is available after two years.

Both the regional forecasts and reanalysis products are generated at hourly temporal resolution and at 0.1° horizontal spatial resolution with vertical levels of 0 m (surface), 50 m, 100 m, 250 m, 500 m, 750 m, 1000 m, 2000 m, 3000 m, 5000 m from the surface. Here, we computed daily averages from the surface level of the ensemble forecast and the ensemble interim reanalysis for the years 2021–2022. Due to data availability for the study period, the interim ensemble reanalysis is considered over validated reanalysis and will be referred to as ‘reanalysis’ from here on. All the CAMS data used in the study are available for download from the Atmospheric Data Store (CAMS, 2024a, 2024b).

2.1.3. VIIRS aerosol optical depth

VIIRS is one of the five instruments aboard the National Oceanic and Atmospheric Administration – 20 (NOAA-20) satellite. The VIIRS instrument is intended to gradually replace MODIS as the latter nears its end of mission operations. It measures radiance at 22 visible and infrared wavelengths, which is used for deriving cloud and aerosol properties, among other applications. NOAA-20, launched in November 2017, is a polar-orbiting satellite providing global coverage with an equator local overpass time of 13:30 local time in the ascending node and a repeat cycle of 16 days. We use data from NOAA-20 as opposed to the VIIRS instrument on the Suomi National Polar-orbiting Partnership satellite because of the former’s longer expected lifetime in orbit for potential operational use of our model. The VIIRS instrument, with its wider swath of 3,000 km compared to MODIS (swath of 2,330 km), is a scanning radiometer intended to continue the heritage of MODIS instruments, (Sayer et al., 2019). Among other products, VIIRS provides AOD information at 550 nm.

Although AOD does not directly quantify $PM_{2.5}$ concentrations, it is an indirect indicator of the amount and type of aerosol particles present in the atmosphere. Ångström Exponent (AE) can be indicative of different aerosol properties, thus relating to different aerosol sources

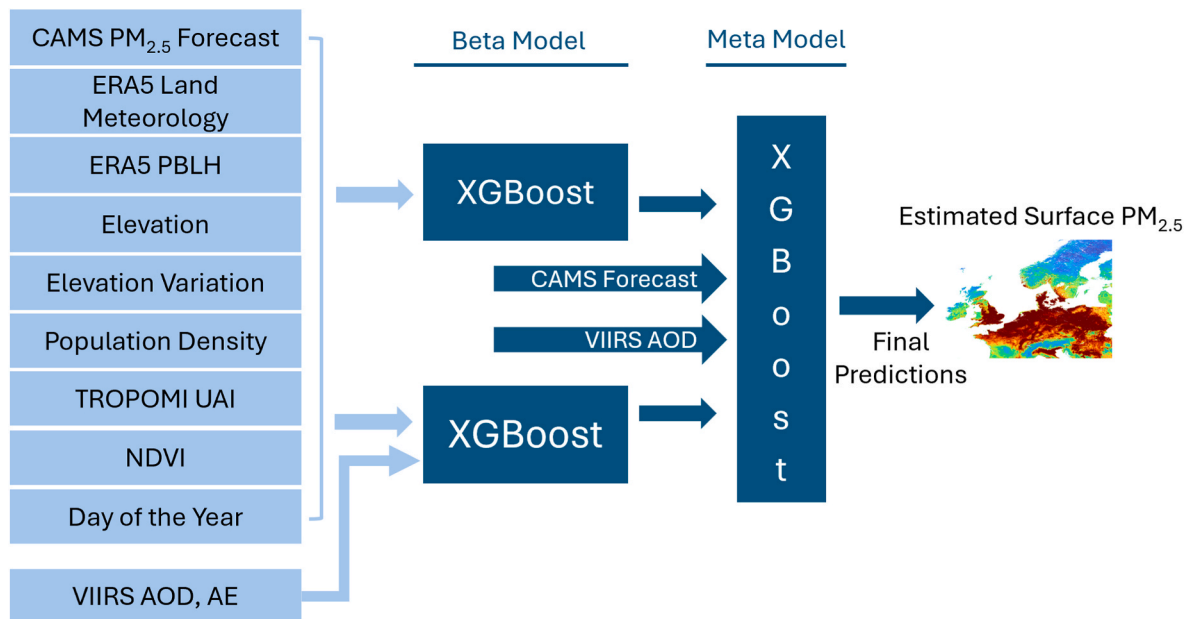


Fig. 1. Input datasets and the concept of the stacked ML approach used in this study. Dew point temperature, 2 m air temperature, surface pressure and total precipitation form the inputs from ERA5-Land meteorology.

(Favez et al., 2009). Our study uses the V2.0 Deep Blue (DB) Level-2 aerosol product (AOD and AE) from VIIRS (C. Hsu, 2018). The DB AOD retrieval algorithm uses blue channels where the surface reflectance is stable over bright surfaces such as urban areas and dynamically estimates surface reflectance over vegetated land surfaces (N. C. Hsu et al., 2019). Version 2 of this product has improvements over version 1 in terms of better smoke detection schemes, and better accounting of surface pressure resulting in improved retrievals over elevated areas (NASA, 2022). The AE product of VIIRS is retrieved at wavelength ratios 412/488 nm or 488/670 nm, depending on the type of land surface, with the former ratio used for arid surfaces and the latter for vegetated/mixed surfaces. The Level-2 products are available at a spatial resolution of 6 km $\times$ 6 km and were downloaded from the LAADS DAAC archive (DAAC, 2023). These Level-2 scientific datasets are generally filtered to exclude clouds, sea ice, and surface snow. Additionally, the data contains a quality assurance flag to identify retrievals with suspected problems. The quality flag has values from 1 to 3 with 1 indicating the worst retrieval, and 3 representing the highest quality. Here only the recommended best estimates over land for AOD and AE are considered excluding pixels with quality flag 1.

2.1.4. ERA5 and ERA5-Land meteorological variables

ERA5 is a reanalysis of weather and climate produced by ECMWF. It is based on the IFS model and uses 4D-Var assimilation of a wide variety of observations (Hersbach et al., 2020, 2023). ERA5 meteorological data are provided at a horizontal spatial resolution of 0.25° with an hourly temporal resolution on a series of pressure levels. Using ERA5 atmospheric variables as an input, the ERA5-Land reanalysis (Muñoz-Sabater et al., 2021) is a replay of the land component of the ERA5 reanalysis and provides hourly surface meteorological variables at a horizontal spatial resolution of 0.1°. Due to the finer spatial resolution, in this study, we primarily used data from ERA5-Land, namely total precipitation, surface pressure, air temperature at 2 m height, and wind information at 10 m height. Planetary Boundary Layer Height (PBLH) is not available from ERA5-Land, so we used PBLH data from ERA5. Both datasets were acquired from the Copernicus Climate Data Store (Climate Data Store, 2024).

2.1.5. Other ancillary datasets/predictors

Other spatial and temporal factors that can affect surface PM_{2.5}

concentration are also taken into account. These include topography, population density, Normalized Difference Vegetation Index (NDVI), aerosol absorption index, and an indicator of the day of the year (Favez et al., 2009; Guan et al., 2021; X. Xu et al., 2021). For topography, elevation and its variation in terms of standard deviation within a 5 km neighborhood are used. Elevation information is taken from the space-based Multi-Error-Removed Improved-Terrain (MERIT) Digital Elevation Model (DEM) datasets available at a spatial resolution of ~90 m (Yamazaki et al., 2017). Population density for the latest available year 2020 from the 'Gridded Population of World Version 4 (GPWv4)' dataset provided by the Socioeconomic Data and Applications Center was selected for this study (CIESIN, 2018; Doxsey-Whitfield et al., 2015). It is available at approximately 1 km spatial resolution. NDVI is a very commonly used satellite-based metric for estimating vegetation health, and at the same time, it is an approximate indicator of land cover types. Land Cover types such as built-up and forest can indirectly relate to PM_{2.5} levels (H. Jin et al., 2022; G. Xu et al., 2020). To keep the impact of seasonal vegetation changes and as a proxy for land cover, in this study, the 16-day Level-3 VIIRS-based NDVI product available at 500 m spatial resolution was used (Didan, K. & Barreto, 2018). The Tropospheric Monitoring Instrument (TROPOMI) on-board the Sentinel-5P satellite provides aerosol-related data in terms of Ultraviolet Aerosol Index (UAI) at a spatial resolution of 3.5 km $\times$ 5.5 km at nadir (Apituley et al., 2018). We selected the reprocessed Level-2 aerosol absorption index dataset from TROPOMI, using the recommended quality assurance filter of values above 0.8. These data were acquired from the Copernicus Data Space Ecosystem (Copernicus, 2024).

2.2. Methodology

2.2.1. Data preprocessing

The input data used in the ML model predictions are available at different spatial resolutions, grids, and projections. These input datasets are selected based on existing studies and the domain knowledge of factors influencing the surface PM_{2.5} concentrations (Tang et al., 2024; X. Xu et al., 2021). After exploratory analysis, and removing features such as relative humidity, and night lights that had low feature ranking and high correlation with other variables, the final set of input variables as shown in Fig. 1 were selected. All the input datasets were re-gridded to a common target grid with 1 km $\times$ 1 km spatial resolution. Datasets

with finer spatial resolution than the target grid size were spatially aggregated to the 1 km grid using the mean of all valid fine-resolution pixels. Coarser-resolution datasets were bilinearly interpolated to match the target grid. In terms of temporal resolution, several datasets including the CAMS PM_{2.5} forecast, the meteorological parameters from ERA5/ERA5-land, and the station PM_{2.5} observations from EEA were available at hourly intervals. As our objective here was to estimate daily average surface PM_{2.5} concentrations, all datasets were averaged to daily temporal resolution. For the CAMS forecast data, a daily average was computed from 24 h of data. Note that for VIIRS AOD and AE, the quality-filtered daily data do not represent daily averages as in the case of model data but represent information from the satellite overpasses that generally cross Europe between 10:20–14:00 UTC and 00:35–4:00 UTC. For training and testing the ML model, values are extracted from all the gridded input datasets at the location of the monitoring sites. In addition, data transformations are applied to the input datasets as described in [Supplementary Table S1](#) to avoid skewness in the data distribution.

2.2.2. ML model fitting

Of the 1,773 ground monitoring stations, 80% of stations were randomly selected and reserved for training and 20% for the evaluation of the ML model (test set). This study uses the XGBoost ML algorithm ([Chen and Guestrin, 2016](#)) to train models for estimating surface PM_{2.5} concentration across Europe. XGBoost is a tree-based ensemble ML algorithm that uses a gradient boosting technique to sequentially build a tree by minimizing the subsequent prediction errors. Due to its structure, it is designed and proven to be computationally efficient, which makes it suitable for large-scale study areas ([Tang et al., 2024](#); [Unik et al., 2023](#); [Wei et al., 2020](#); [Q. Xiao et al., 2018](#)).

Some of our input datasets are subject to spatial data gaps due to cloud contamination or other retrieval issues (e.g., VIIRS AOD, VIIRS AE). Such gaps would in principle limit the prediction of surface PM_{2.5} to those areas where these datasets provide measurements. In order to address this limitation and to make predictions for the entire domain, while taking advantage of other spatially continuous input datasets, we use a stacked ML approach, which is an ensemble ML algorithm working in two stages. The first stage contains a group of two or more ML models called Beta models that use some input datasets to provide a prediction. The second stage contains a single model called the Meta model whose inputs are predictions from the Beta models. The Meta model can additionally also take other input variables based on the requirements. In a way, the Meta model acts as a bias corrector. This ensemble approach helps in best combining the predictions from multiple models to provide an overall better-performing model compared to singular models.

Here we employ a stacked ML approach to combine predictions from two XGBoost-based models, each utilizing slightly different input datasets, as shown in [Fig. 1](#) to get a spatially continuous and smooth output. Our Beta model (first stage) comprises of two XGBoost models: one trained with all the input datasets, including VIIRS AOD and AE, and the other trained with all input datasets except AOD and AE. The Meta model (second stage) is an XGBoost model that takes four inputs: the predictions from the two Beta models, along with VIIRS AOD and the CAMS PM_{2.5} forecast. The latter two are additional inputs to the Meta model, alongside the two Beta models. They allow the Meta model to differentiate between scenarios where AOD data is available and those where it is missing, and as such which Beta model is more relevant at a given location. This stacking approach ensures that our trained ML model can estimate surface PM_{2.5} even when AOD values are missing, in this case using the information available from the CAMS forecast and the other input parameters. This ensemble approach in our case mainly helps address spatial gaps and artifacts from satellite AOD while also effectively combining predictions from multiple models to achieve better or similar overall performance than individual models. To prevent overfitting or data leakage, the beta and meta models are trained using

different sets of training data. Hence, from 80% of the stations selected for training, 60% are randomly reserved for training the beta model and 40% for the meta model. The meta model is therefore trained using the predictions from the Beta model with 40% reserved station data as input. [Supplementary Fig. S1](#) shows the performance of the two Beta models and their advantage in generating spatially continuous data. The hyperparameters used in the XGBoost models are shown in [Supplementary Table S2](#).

2.2.3. Evaluation and comparison

The model performance is evaluated using 20% of the station data that was not utilized during the training process. Uncertainty indicators such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Bias (MB), Spearman Correlation (R), and Coefficient of Determination (R²) are employed to evaluate the statistical performance of the model. In addition, the Relative Absolute Error (RAE) in percentage is used as it provides a quantification of normalized error ([Equation \(1\)](#)). To assess how closely S-MESH estimations align with the CAMS reanalysis in terms of performance and spatial patterns, a comparison with the CAMS reanalysis is also carried out.

$$RAE = \frac{abs(Model\ Estimation - Station\ Observation)}{Station\ Observation} * 100 \quad 1$$

The importance of the input variables in the stacked model is evaluated based on SHapley Additive exPlanations (SHAP) value importance on the test dataset. This approach derives the importance of an input variable based on its interaction with other variables and its influence on model predictions ([Lundberg et al., 2020](#)). Such an approach is suitable as it considers the impact of an input variable on the predictions rather than only the change in accuracy ([Lundberg and Lee, 2017](#)). To evaluate feature importance of the stacked S-MESH model, SHAP values are first calculated independently for each model in the stacked framework (two beta models and one meta-model). This step captures the influence of each feature within the beta and meta models. Due to SHAP's additive property and the varying contribution of each beta model to the final predictions, a weighted average of the SHAP values is used to calculate the overall feature importance for the stacked model. The SHAP values of beta model predictions in the meta-model indicate the influence of beta models on the final output and thus serve as weights in the weighted averaging process.

2.2.4. Computational performance

The S-MESH model for daily PM_{2.5} estimations is trained using the XGBoost algorithm for both the Beta and Meta stages. The choice of the XGBoost method was based on its computational efficiency and prediction capability. The S-MESH model requires ca. 11 s for training on 0.6 million data points with 16 input variables on a system equipped with a 10-core Intel Core i7 1.7 GHz processor and 32 GB RAM. The daily predictions as continental-scale raster maps were performed on a small computing cluster. To generate a single day's estimate for the entire study area, consisting of 37.1 million grid cells, the S-MESH model requires approximately 40 s using 2 CPUs with 14 cores each and 150 GB of memory. The complete data flow including reading the input datasets, preprocessing, predictions, and raster map generation, requires ca. 8 min to produce the prediction for a single day. Daily predictions using this approach were generated for the years 2021 and 2022 and were in addition averaged to generate monthly and annual estimates.

3. Results and discussion

In this section, we demonstrate the model performance and present analyses of the annual and daily PM_{2.5} estimates from S-MESH for the years 2021–2022. The model predictions are evaluated against station measurements and a comparative analysis is carried out against the corresponding CAMS regional forecast and regional reanalysis products. While station observations are used as a standard reference, the CAMS

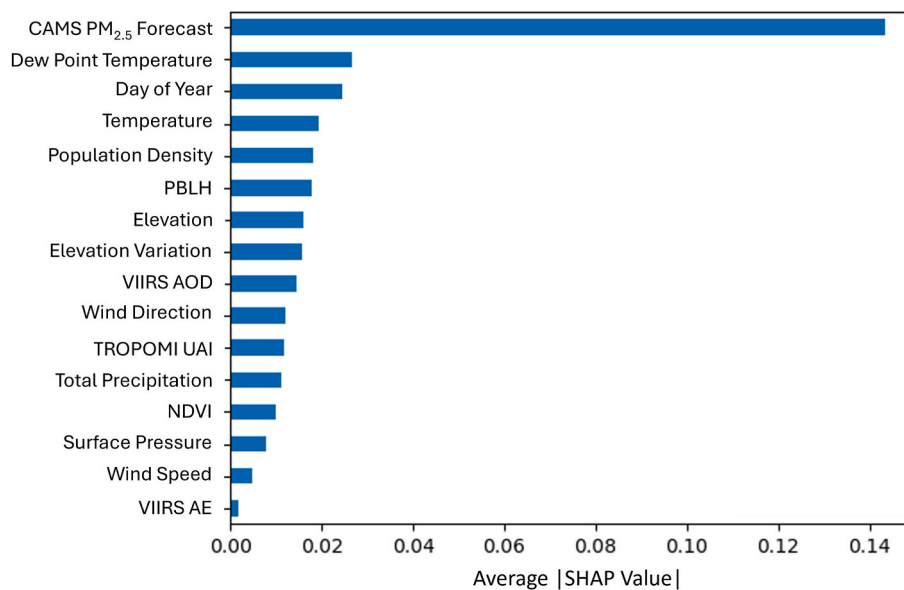


Fig. 2. Importance of the input variables in the S-MESH model estimated by SHAP. The x-axis represents the average SHAP value of inputs on the test data.

reanalysis is used as a reference for assessing the spatial patterns. Finally, to evaluate S-MESH's capability during high pollution events, a high PM_{2.5} pollution episode from March 2022, caused by multiple sources, is analyzed. Here, we identify PM_{2.5} concentrations above the EU daily threshold of $\geq 25 \mu\text{g}/\text{m}^3$ as a high pollution event and episodes as events of elevated levels of PM_{2.5} that impact large areas, lasting for at least 3 days. This can be caused by natural or anthropogenic emissions combined with meteorology. This definition of episode is similar to that described in the CAMS interim report (Tarrasón et al., 2022).

3.1. Input variable importance

Fig. 2 highlights the importance of the input variables for the S-MESH model, derived based on the weighted average of absolute SHAP values for all input variables on the test data. As expected, given the study's objective of downscaling the regional CAMS forecast, the ranking indicates that PM_{2.5} from the CAMS forecast holds the highest importance in the model, contributing approximately 40% to the estimation of PM_{2.5} in S-MESH. Using the CAMS forecast as one of the inputs ensures that physically and chemically resolved concentrations form the basis of S-MESH estimates. The individual contributions from the remaining 15 variables are $<8\%$. Among them, meteorological parameters such as dew point temperature and air temperature at 2 m height are ranked high. The temporal indicator, 'Day of the Year', also exhibits similar importance. The comparatively higher ranking of these variables indicates the influence of temperature, humidity and seasonal variation in estimating PM_{2.5}. These are followed by the static variable population density – that indirectly indicates human activities and emissions and, meteorological variable PBLH – which influences the vertical mixing of the pollutant, with each accounting for ca. 5% of the total importance. Elevation-related variables that help capture the influence of topography are ranked moderately, accounting for ca. 4.5% of the importance. These preceding variable rankings indicate that the plain spatial (population density, elevation) and temporal variables (temporal variables) have similar influence and importance in surface PM_{2.5} estimation as the other spatial-temporal variables. VIIRS AOD occurs 9th in the variable importance ranking and accounts for a total importance contribution to the predictions of ca. 4%. AOD, being an indirect indicator of aerosols in the atmosphere, was expected to rank higher. But since the satellite AOD represents the total columnar particles, it may not always reflect the surface PM_{2.5} variations, resulting in overall moderate contribution compared to some meteorological variables.

Variables such as wind speed and AE rank the lowest, with AE's overall importance being very low. Since AE can act as a complimentary indicator of aerosol properties, this variable was nonetheless retained in the study (J. Jin et al., 2023; P. Shao et al., 2017; Zang et al., 2021).

3.2. Annual PM_{2.5} estimates

Fig. 3 shows annual average PM_{2.5} maps derived from daily estimates over Europe for the years 2021 (left column) and 2022 (right column) as obtained from the CAMS regional forecast (top row Fig. 3a–b), S-MESH (middle row Fig. 3c–d) and the CAMS regional reanalysis (bottom row Fig. 3e–f). Throughout both years of the study, all three models consistently showed high annual average PM_{2.5} concentrations of $>25 \mu\text{g}/\text{m}^3$ over the Po Valley of Northern Italy. Similarly, over Poland, such levels were estimated in the case of S-MESH and the reanalysis. These concentrations surpass the revised annual thresholds as defined by the WHO and EU. More moderate concentrations of $10\text{--}17 \mu\text{g}/\text{m}^3$ are estimated over central and western Europe, and parts of the Iberian Peninsula.

Similar spatial patterns are observed across all model averages except for Eastern Europe. Notably in Poland, Hungary, and Slovakia, S-MESH and the CAMS reanalysis show higher annual average PM_{2.5} surface concentrations for both years compared to the CAMS regional forecast. The overlaid annual averages from station observations also show the presence of high PM_{2.5} levels in these regions and S-MESH, as well as the reanalysis, match those more closely than the forecast (also see Supplementary Fig. S2). This indicates that the S-MESH model, with CAMS PM_{2.5} forecast as the main input, can capture additional information beyond what the forecast provides and align well with the spatial patterns of the reanalysis. Moreover, the error rates of annual estimates of the CAMS forecast, S-MESH, and reanalysis (Supplementary Table S3) highlight the superior performance of S-MESH compared to CAMS.

The estimates from the CAMS reanalysis are generated by assimilating the station observations and thus can be considered a reference for spatial patterns. The spatial patterns, hotspots, and overall values from annual estimates produced by the S-MESH approach closely match those of the CAMS reanalysis. More variations in spatial patterns can be observed in S-MESH (e.g., over Poland, Hungary, Romania, France, or Italy). However, over northern Europe, S-MESH exhibits slightly higher values than both the forecast and reanalysis. This slight positive bias from S-MESH, however, closely matches the annual average PM_{2.5} concentrations measured by the monitoring stations in this region, as

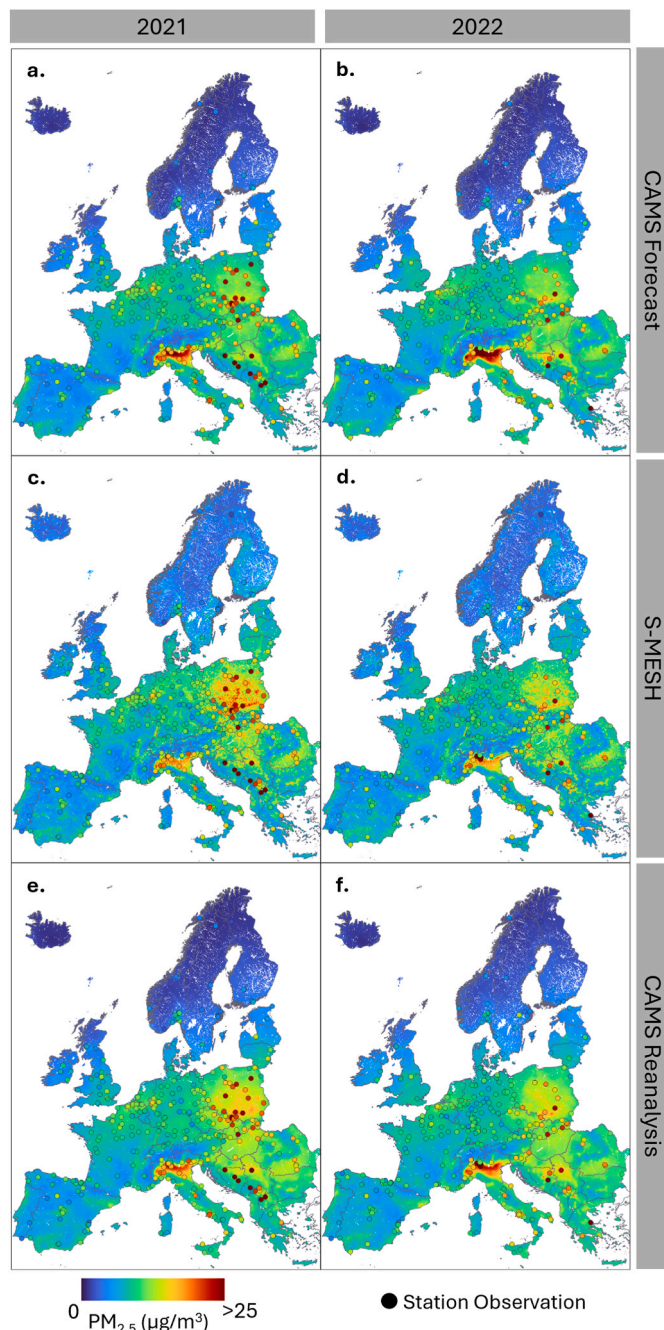


Fig. 3. Annual average surface $PM_{2.5}$ estimates in units of $\mu g/m^3$ for the years 2021 (left column) and 2022 (right column) derived from the CAMS regional forecast (top row), S-MESH (middle row) and the CAMS regional reanalysis (bottom row). The test station observations, showing the measured annual average $PM_{2.5}$ concentration at each site, are overlaid using the same color scale as the maps.

seen by the overlaid station observations in Fig. 3. Supplementary Fig. S2 further shows these improvements in terms of annual relative errors. In contrast, the S-MESH annual average $PM_{2.5}$ estimates over Northern Italy exhibit slightly lower values compared to the forecast. More in-depth analysis shows that these somewhat reduced estimates shown by both S-MESH and the reanalysis more closely match the annual averages from station observations.

In order to demonstrate the improved spatial detail provided by S-MESH compared to the reanalysis, Fig. 4 shows local-scale maps of 2021 annual average $PM_{2.5}$ of S-MESH and the reanalysis for an arbitrarily

selected relatively small region in Italy, with both models shown at original spatial resolutions of ~ 1 km and ~ 10 km respectively. As can be seen in this figure, due to its higher spatial resolution, S-MESH better captures spatial variability of $PM_{2.5}$ over local scales (Fig. 4a), which cannot be adequately represented at the coarser spatial resolution provided by the CAMS reanalysis (Fig. 4b). Such detailed spatial patterns provide valuable information for local-scale air quality assessments. Quantitative evaluation of such high-resolution local-scale spatial patterns is challenging as typically no reliable spatial reference dataset is available at this spatial resolution, but a comparison against station observations gives some indication of their reliability.

3.3. Daily $PM_{2.5}$ estimates

Daily concentration maps were produced using S-MESH for all days in the 2021–2022 study period. Examples of these daily $PM_{2.5}$ concentration maps can be found in Figs. 9 and 10, Supplementary Figs. S3 and S4, and these will be discussed later in more detail in Section 3.4. Here we initially evaluate and discuss the performance of the S-MESH-based maps. Fig. 5 shows a quantitative evaluation of the daily S-MESH predictions in comparison to the CAMS forecast and reanalysis. More specifically, scatter plots visualize the daily average $PM_{2.5}$ estimates from the CAMS forecast (Fig. 5a), S-MESH (Fig. 5b), and the CAMS reanalysis (Fig. 5c) against daily averaged $PM_{2.5}$ concentration from the station measurements that were reserved for testing.

The summary statistics from the scatter plots show that S-MESH performs quite well for surface $PM_{2.5}$ estimates with a MB of $-0.3 \mu g/m^3$, a MAE of $3.54 \mu g/m^3$, and an RMSE of $6.13 \mu g/m^3$. As such, S-MESH exhibits a significantly lower bias than both the forecast and the reanalysis. In addition, it shows slightly reduced RMSE with respect to the forecast and the reanalysis, respectively. In terms of MAE, the reanalysis is the best performing model, followed by S-MESH and the forecast. The Spearman correlation R for the forecast is 0.72, while S-MESH and reanalysis have a similar R of 0.79 and 0.82 respectively. The R^2 of the S-MESH model is 0.58, which is slightly higher than that of the reanalysis with 0.56 and significantly higher than that of the forecast with a value of 0.39. The summary statistics shown for S-MESH indicate that the approach is capable of producing daily average estimates of surface $PM_{2.5}$ at an accuracy level that outperforms the forecast in all metrics and is very close to and to some extent (for R^2 , MB, and RMSE) surpasses that of the CAMS reanalysis.

The main increase in performance of the S-MESH model over the reanalysis can be found at moderate to high concentration levels. Fig. 5d shows the RAEs of the different models at different $PM_{2.5}$ concentration levels. The estimates of the S-MESH approach exhibit a better performance or lower RAE compared to the CAMS reanalysis for concentrations $>20 \mu g/m^3$, while the reanalysis has lower RAE with respect to S-MESH at concentrations $<20 \mu g/m^3$. However, in terms of absolute error measures such as Bias, the error rates at concentrations $<20 \mu g/m^3$ are still low and comparable to CAMS reanalysis as seen in Supplementary Fig. S5. At concentrations $>20 \mu g/m^3$, the MAE and MB of S-MESH are at $10 \mu g/m^3$ and $-7.3 \mu g/m^3$ respectively, while that of reanalysis is at $13.3 \mu g/m^3$ and $-11 \mu g/m^3$ respectively. While the RAE of S-MESH is lower than that of the forecast across most of the concentration range, it remains similar to that of the forecast at concentrations $<10 \mu g/m^3$. This suggests that under very clean conditions, the predictor variables used in S-MESH are not able to significantly enhance the CAMS forecast but the bias distribution in Supplementary Fig. S5 suggests that the S-MESH approach decreases the negative bias in the forecast. When considering S-MESH primarily as a downscaling method for CAMS, it further demonstrates that while downscaling may increase spatial detail, it does not add much value in clean conditions, as the spatial variability in the concentration field is then typically minimal and spatial gradients are weak.

Fig. 5e shows the RAEs across station types — Background, Traffic, and Industry. S-MESH has a median RAE ranging from 23% to 27% with

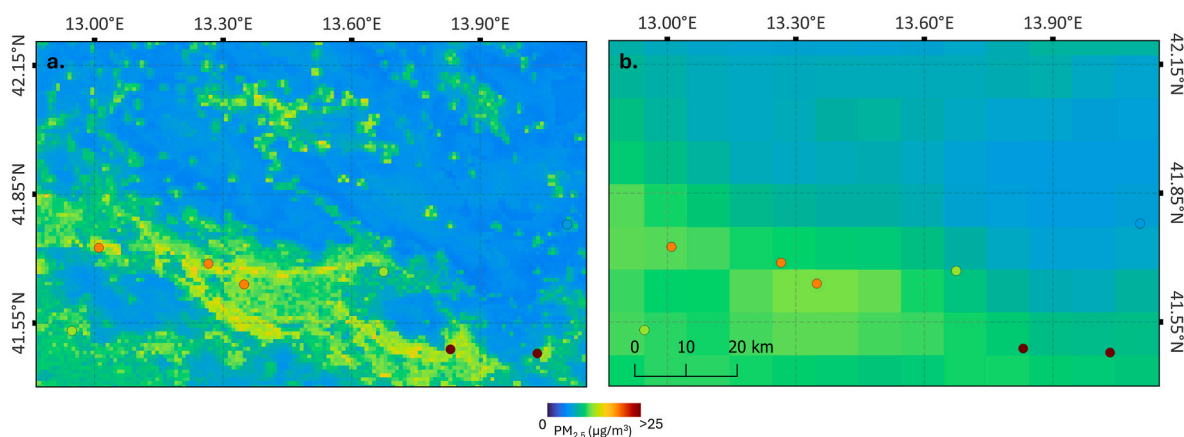


Fig. 4. An example of “zoomed-in” maps of the 2021 annual average $PM_{2.5}$ concentration for a relatively small region in south-west Italy from (a) S-MESH and (b) the CAMS reanalysis at their respective original spatial resolutions of ~ 1 km and ~ 10 km. Overlaid are the annual averages from station observations in the same color scale as the map.

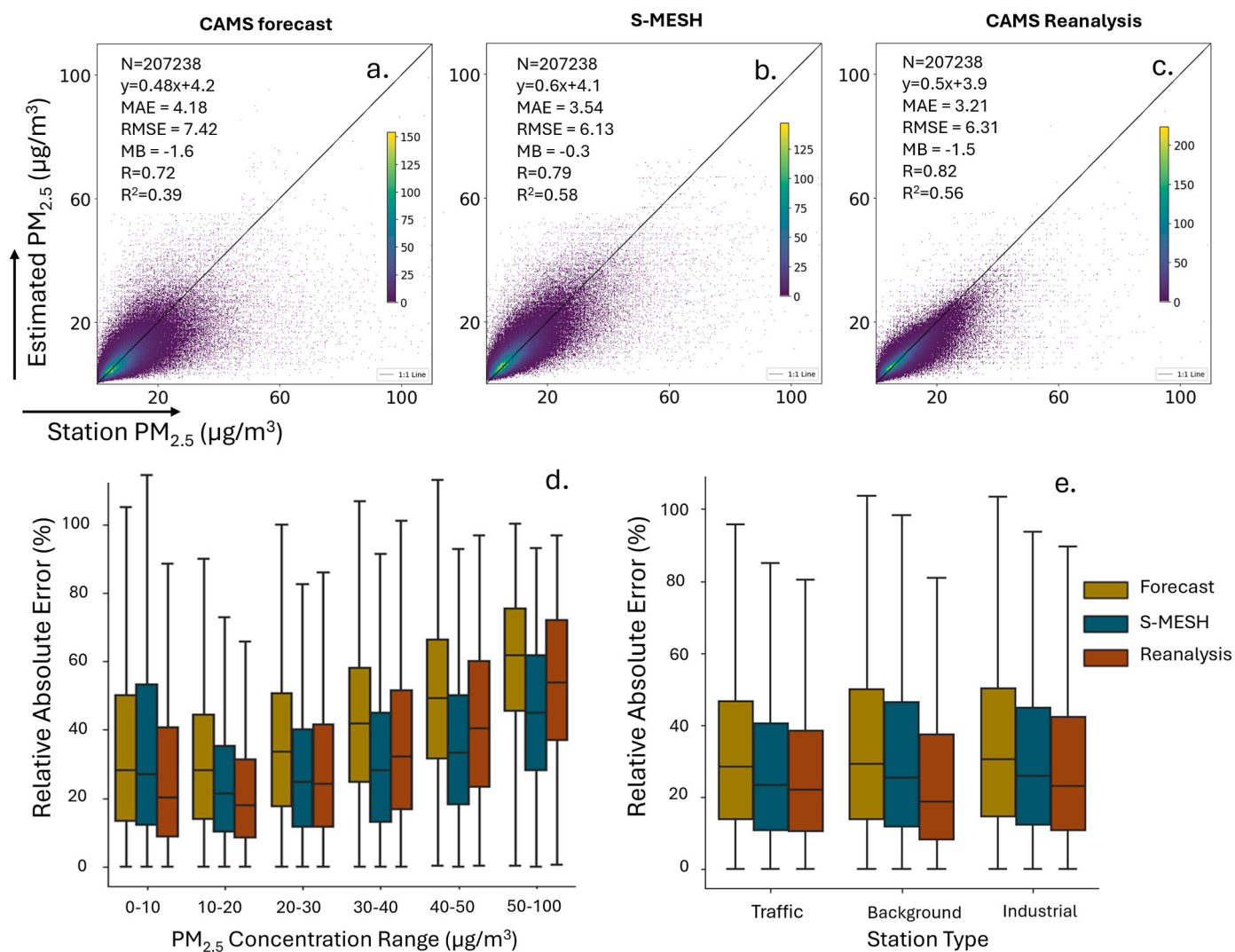


Fig. 5. Evaluation of CAMS regional forecast, S-MESH, and CAMS reanalysis daily predictions for 2021–2022 against the test stations' daily observations. The first row shows scatter plots of model predictions against station observations for (a) CAMS forecast (b) S-MESH and (c) CAMS reanalysis. MAE, RMSE, and MB are given in $\mu g/m^3$. The second row contains box plots (with 25th, 50th, 75th percentiles) of relative absolute errors (in percent) for all three approaches as a function of (d) concentration level and (e) air quality station type.

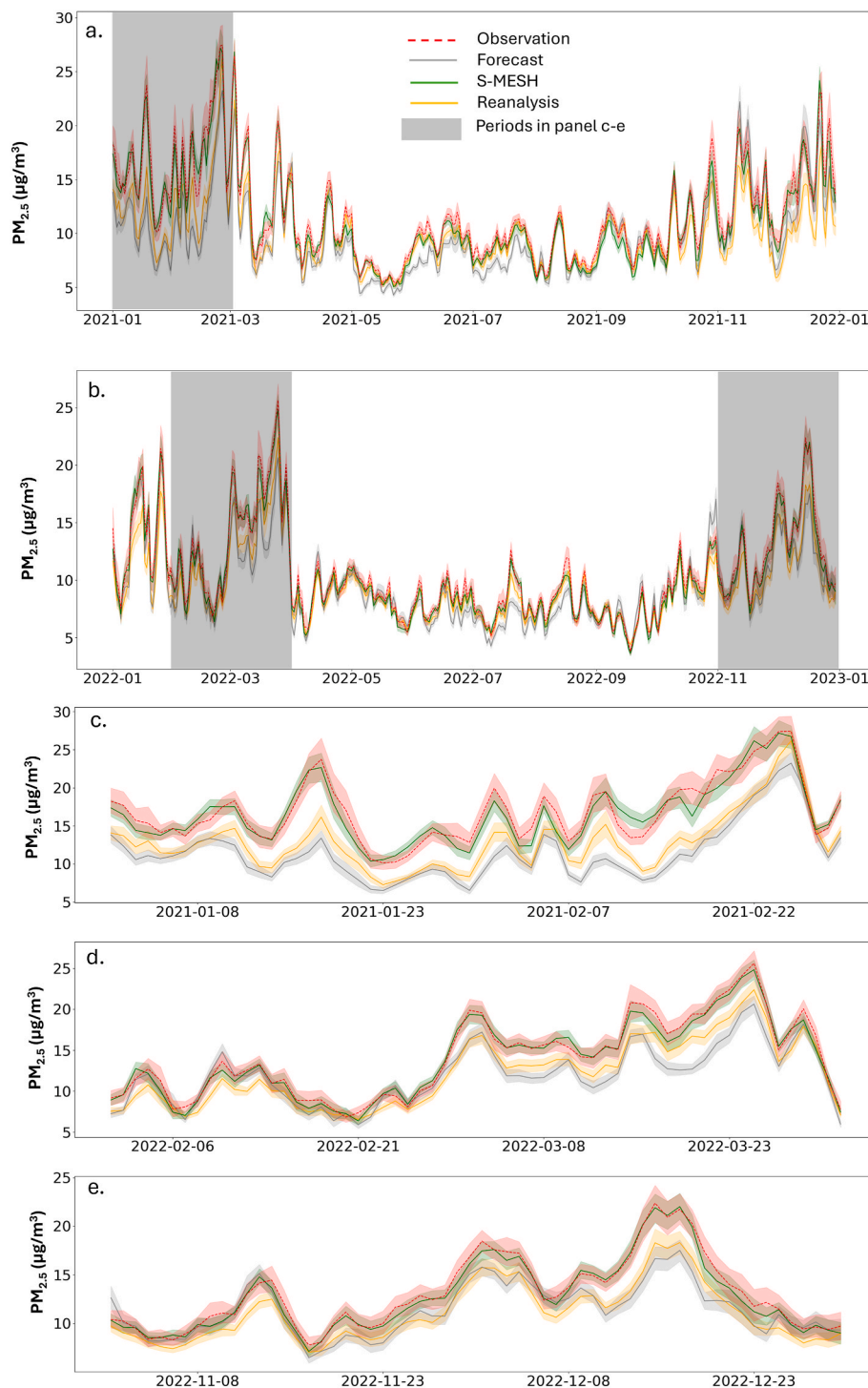


Fig. 6. Time series of daily average PM_{2.5} concentration averaged over all test stations in the study domain for the years 2021 (a) and 2022 (b) and derived from CAMS forecast (grey), S-MESH (green), the CAMS reanalysis (orange), and station observations (red). Shades around the lines show the 95% confidence intervals of daily average values from test stations. For improved clarity, panels c–e show a few periods during high pollution and episodes in more detail.

better performance over the more polluted traffic regions. The RAE of S-MESH is lower compared to the forecast and is comparable with the reanalysis except over background regions where the reanalysis has a lower median RAE of 18%. This relates to the performance of the three models at low concentration conditions which are typically associated with background stations.

S-MESH is trained and tested on a dataset containing scenarios with and without gaps in AOD data. Given that AOD is significantly influenced by clouds, a higher percentage of the data features AOD gaps.

Nevertheless, the stacked design of the beta models - where one model incorporates satellite AOD as input while the other does not - effectively addresses such scenarios, as illustrated in [Supplementary Fig. S6](#). It shows that in both scenarios, the S-MESH model performs similarly with slightly better MAE and MB when AOD data is not available (MAE = 3.48 µg/m³, MB = −0.73 µg/m³ without AOD and MAE = 3.81 µg/m³, MB = −0.45 µg/m³ when AOD is available). This is attributed to the larger datasets representing this scenario during training. On the other hand, a slight improvement in terms of R (0.78 without AOD vs 0.8 with

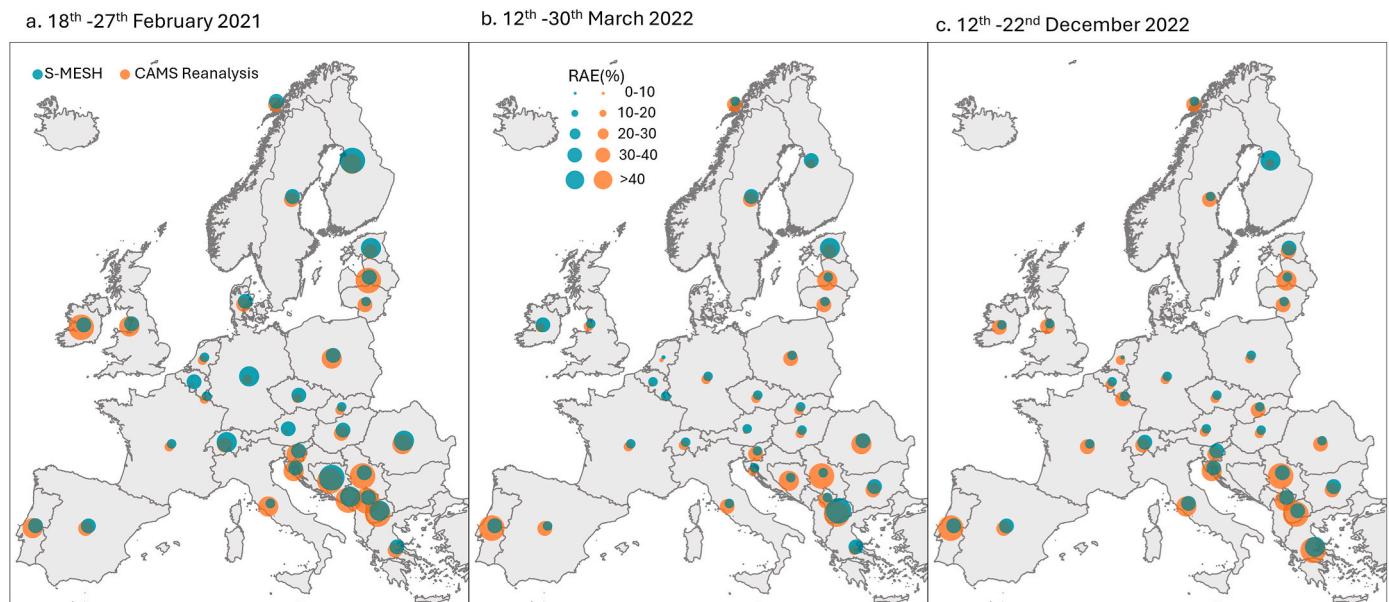


Fig. 7. Country-wise median RAE (in percent) of S-MESH (in green) and the CAMS reanalysis (in orange) for episodes during the periods (a) 18th-27th February 2021 (b) 11th-30th March 2022 (c) 12th-22nd December 2022. The size of the circles represents the RAE with a smaller size indicating lower RAE and hence better performance. The location of the symbols is shifted slightly between the two models for clarity. Due to a lack of suitable station observations, an evaluation over Iceland is not performed.

AOD) and R^2 (0.57 without AOD vs 0.6 with AOD) is observed when AOD data is available, indicating that the AOD data is useful in explaining the $PM_{2.5}$ variations but does not significantly change the results when missing.

Fig. 6 shows the time series of the daily average $PM_{2.5}$ estimates for the years 2021–2022 derived by averaging over the locations of the test stations the estimates from the CAMS regional forecast, S-MESH, the reanalysis, and the actual station observations. While Fig. 6a and b contain the whole time series for the years 2021 and 2022, respectively, Fig. 6c through Fig. 6e shows shorter time series around a few periods of high $PM_{2.5}$ levels and/or episodes for improved clarity. All three models capture the temporal variation well. This day-to-day temporal variation of estimates has a Spearman correlation of 0.87 for the forecast, 0.97 for S-MESH, 0.92 for the reanalysis as well as an RAE of 15%, 5%, and 10% respectively. These metrics together with Fig. 6 indicate that the S-MESH and the reanalysis follow the station observations more closely than the forecast. In addition, S-MESH provides significantly better estimates of the daily temporal variability than the reanalysis as demonstrated by an RAE that is more than half that of the reanalysis.

High $PM_{2.5}$ concentrations can be observed in the time-series during the winter months of December and January in both years. These are also the periods that are most influenced by the presence of clouds resulting in large AOD gaps. Despite this, the S-MESH model predictions closely follow the station observations as seen in Fig. 6 and further indicate the model's robustness in these scenarios. During January 2021, the average $PM_{2.5}$ estimate (with standard deviation) is $15 \pm 14 \mu g/m^3$ from station observations, $15 \pm 9 \mu g/m^3$ from S-MESH, $11 \pm 8 \mu g/m^3$ from the reanalysis, and $10 \pm 7 \mu g/m^3$ from the forecast. $PM_{2.5}$ levels are slightly lower during January 2022 with average concentrations of $13 \pm 11 \mu g/m^3$, $13 \pm 9 \mu g/m^3$, $11 \pm 7 \mu g/m^3$ and $12 \pm 9 \mu g/m^3$ from stations, S-MESH, reanalysis, and forecast, respectively.

Additionally, during February 2021 (Fig. 6c), March 2022 (Fig. 6d) and December 2022 (Fig. 6e) cases with continuous peaks of high pollution days are observed. These are specific pollution episodes that had impacts over most of Europe. As seen from the overlap of red and green lines in Fig. 6c, d and e, predictions and temporal variations during these periods are better captured by the S-MESH model than the reanalysis. This indicates that S-MESH better captures high

concentration levels compared to CAMS and hence can be a reliable source for analyzing high pollution events. In the following section, we analyze one of these events in more detail to better understand how the S-MESH model behaves for different sources of PM pollution.

Fig. 7 shows the country-wise median RAE of S-MESH and the reanalysis of the above mentioned episodes which more specifically occurred for the periods 18th-27th February 2021 (Fig. 7a) for February 2021 episode (Tarrasón et al., 2022), 12th-30th March 2022 (Fig. 7b) for the March 2022 episode and 12th-22nd December 2022 (Fig. 7c) for December episode (Hamer et al., 2023). The March 2022 episode occurred in two phases and only the latter and more severe of the two is considered here for evaluation. As can be seen in Fig. 7, across most countries during these episodes, S-MESH exhibits lower or similar RAE (indicated through a smaller circle size in green) compared to the reanalysis. Notably, over Poland, Romania, Latvia, Lithuania, Italy, and southeastern Europe, significant improvements in S-MESH over the reanalysis can be observed. Better performance can also be seen in Portugal. Over regions such as Finland, which is mainly dominated by low concentration levels during this period, S-MESH's performance is poorer than that of the reanalysis.

Although S-MESH generally demonstrates improved performance at concentrations above $20 \mu g/m^3$, these enhancements are particularly pronounced when compared to reanalysis data over Poland, Hungary, Romania, and parts of Eastern Europe, as well as Italy and the south-eastern European regions. Supplementary Table S4 shows the country-wise performance of model estimates during the whole study period of 2021–2022 over all test data and cases where daily station observations of test data are high (here $\geq 25 \mu g/m^3$ considering the EU daily thresholds). While a general improvement compared to the forecast and similar performance with respect to the reanalysis can be seen, differences are observed over countries such as Finland and Portugal, where S-MESH has higher relative error compared to forecast. This is because these regions are dominated by average low $PM_{2.5}$ concentrations of $5 \mu g/m^3$ which S-MESH fails to capture leading to higher relative errors. As seen before, S-MESH overestimates very low concentrations. One way to potentially improve predictions at very low concentration levels, particularly at $< 5 \mu g/m^3$ would be to use other loss functions in XGBoost such as Quantile loss or weighted loss (Sluijterman et al., 2024)

Table 1

Overall spatial and temporal statistical indicators of CAMS forecast, S-MESH and CAMS reanalysis estimates over test stations.

	Overall			Spatial			Temporal		
	RAE (%)	R	R ²	RAE (%)	R	R ²	RAE (%)	R	R ²
Forecast	29	0.72	0.39	36	0.71	0.35	15	0.87	0.67
S-MESH	24	0.79	0.58	32	0.81	0.52	5	0.97	0.96
Reanalysis	20	0.82	0.56	28	0.86	0.62	10	0.92	0.78

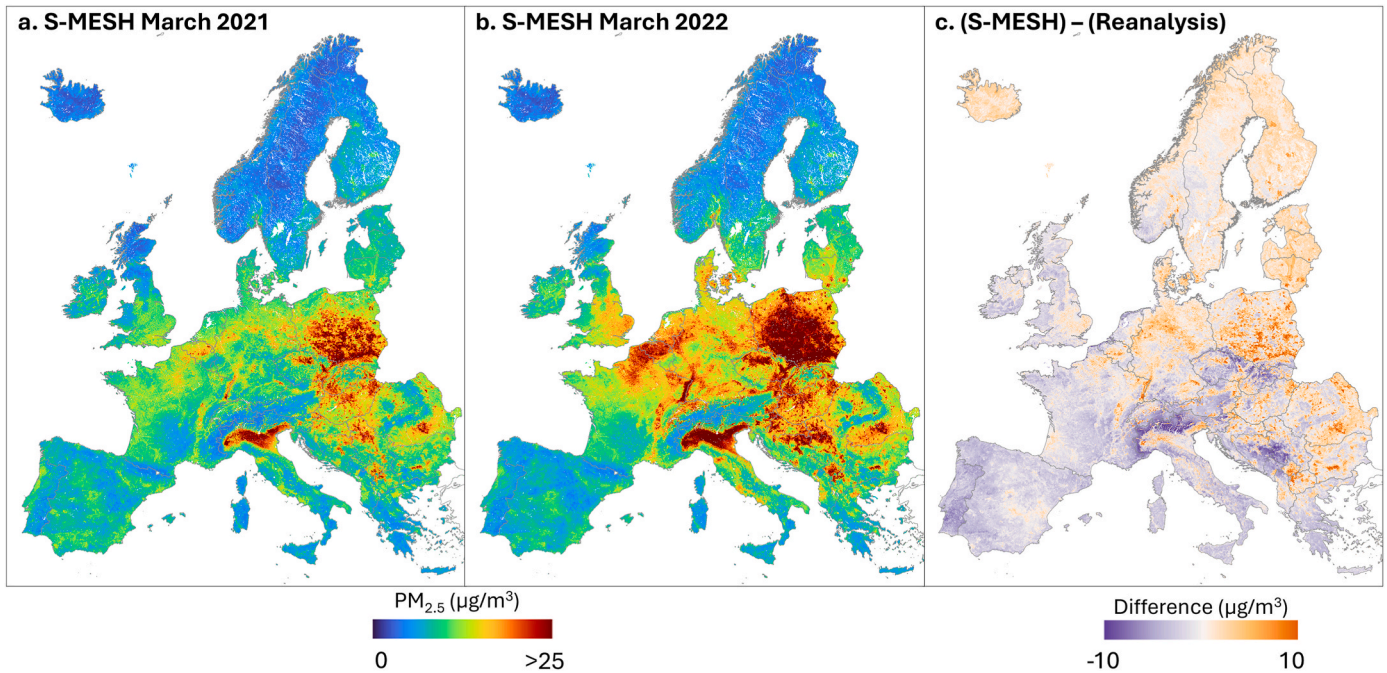


Fig. 8. S-MESH-based $PM_{2.5}$ averages (in $\mu g/m^3$) for (a) March 2021, (b) March 2022, as well as (c) S-MESH minus reanalysis for March 2022.

that allow to differently penalize the under and over predicted values during the training process, unlike the default mean squared error. As such, the overall spatial accuracy of S-MESH in terms of RAE is slightly lower at 32% than the reanalysis at 28%. However, S-MESH shows a clear improvement over the forecast (36%) as presented in Table 1, which summarizes and illustrates the overall spatial and temporal RAE, R and R^2 for estimates from the three models. The temporal statistical indicators depict superior performance of the S-MESH model in the temporal dimension and improvements compared to both the forecast and the reanalysis in all aspects. To further understand the spatial performance, we examine the S-MESH model performance as a function of training station density (Supplementary Fig. S7). Increasing the station density leads to a decrease in RAE when the number of stations is less than 7 within a space of $0.08^\circ \times 0.08^\circ$ grid, with the reduction becoming more gradual between 7 and 10 stations. Beyond this range, the decrease in RAE is not significant. These are indicative that the current S-MESH model generalizes well to unseen data and is not biased to a specific station/region in the study area. Additionally, with an available average training station density of 10 stations per 0.08° grid, the current S-MESH model optimizes well for these conditions.

3.4. $PM_{2.5}$ episode during March 2022

In March 2022, a $PM_{2.5}$ episode as seen in Fig. 6e occurred over Europe that developed in two phases from March 1st-10th and March 12th-30th (Hamer et al., 2023). This episode resulted from the combined effect of different sources and led to exceedances of the $PM_{2.5}$ levels beyond the EU daily threshold of $25 \mu g/m^3$. Fig. 8 shows the impact of this episode on monthly means. For comparison, Fig. 8a shows the

monthly mean for March from S-MESH during a normal period in March 2021, whereas Fig. 8b shows the situation for March 2022. The impact of the episode in 2022 can be observed in the S-MESH output as widespread very high $PM_{2.5}$ concentrations in March 2022 (data distribution in Supplementary Fig. S8 showing more frequency of $>20 \mu g/m^3$ in 2022 compared to 2021). Very high $PM_{2.5}$ concentrations can be observed in March 2022 over Eastern Europe, parts of France, Germany, Denmark, Austria, Netherlands, Belgium, Luxembourg, and northern parts of Italy. The southern parts of the United Kingdom, Denmark, and Estonia also show elevated average concentrations with values of more than $15 \mu g/m^3$.

Fig. 8c shows the difference between the March 2022 monthly means estimated by S-MESH and the reanalysis. Higher predictions than the reanalysis are estimated by S-MESH over most of Northern Europe, central Europe such as parts of Germany, most of Poland, parts of Hungary, Slovakia, and eastern Europe, whereas the opposite can be seen over Portugal and parts of Italy. Fig. 9 shows the daily $PM_{2.5}$ levels as estimated by S-MESH and the reanalysis during the period of 22nd-26th March 2022. The second phase of this episode was more severe (Hamer et al., 2023). The impact of the second phase on $PM_{2.5}$ levels and its extent over Europe can be seen on comparing the Supplementary Figs. S3 and S4 that show daily average S-MESH $PM_{2.5}$ estimates between 12 and 30th March during a normal period in 2021 and a episode period of 2022, respectively. In 2022, very high $PM_{2.5}$ levels remain over Poland and parts of eastern Europe, northern Germany, Denmark, Netherlands, Belgium, Luxembourg, and northern parts of Italy and southern parts of United Kingdom during this period. Throughout this episode and as also seen in Fig. 8c, the S-MESH output shows comparatively higher concentrations than the reanalysis over large areas of

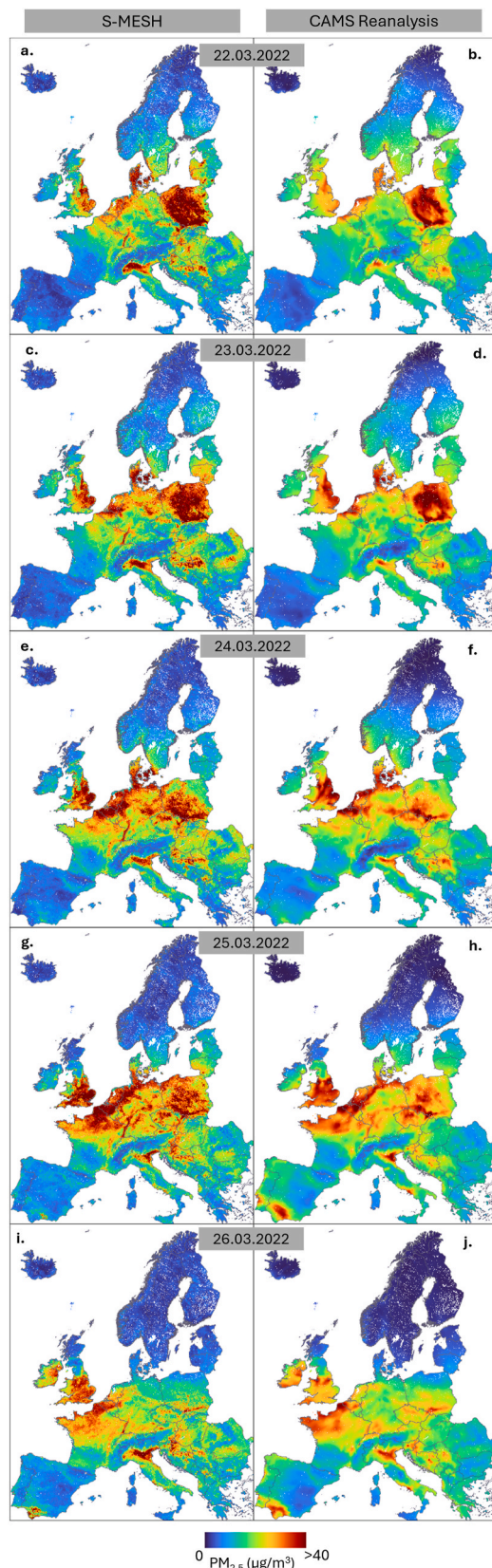


Fig. 9. Daily estimates of surface $PM_{2.5}$ from S-MESH (left column) and the reanalysis (right column) from 22nd–26th March 2022 showing the temporal and spatial variations of $PM_{2.5}$ levels.

Poland, Slovakia, Hungary, and Eastern Europe. Differences in spatial patterns are also observed in Italy, Spain, and parts of Scandinavia. In the following, we analyze a few of these differences with particular focus on Poland, Hungary, Slovakia, and Eastern Europe such as Romania, where S-MESH consistently estimates higher concentrations compared to the reanalysis. Fig. 10 analyzes a few such instances during these periods in more detail to understand the comparative performance of S-MESH and the CAMS reanalysis during the high pollution events. These instances are indicative of the main sources of $PM_{2.5}$ pollution and can be considered representative of the general model performances in these regions during high pollution levels.

Fig. 10a shows $PM_{2.5}$ estimates from S-MESH and the reanalysis over Romania and neighboring countries on March 23, 2022. For better visualization of the details, the forecast is not shown in this panel. High concentrations of $PM_{2.5} > 40 \mu g/m^3$ can be seen in both models, especially over Poland. The high levels of $PM_{2.5}$ from S-MESH show more spatial variation and are observed at more locations than in the reanalysis, especially over Slovakia, Hungary, Romania, Croatia, and Serbia. Some of these are highlighted through the black boxes in Fig. 10a. These spatial variations from S-MESH more closely follow the station observation as seen in Fig. 10a and Supplementary Table S5. The average RAE of S-MESH during this day over Romania is 29% while the RAE of the reanalysis is 50%. During the whole March 2022 pollution episode, the RAE of S-MESH over Romania is 27% and 38% for reanalysis when validated against test station observations. Furthermore, S-MESH shows similar patterns of higher levels and spread of $PM_{2.5}$ over these regions throughout the event.

Fig. 10b shows the $PM_{2.5}$ estimates for the following day, i.e., March 24, 2022 from the CAMS forecast, S-MESH, and the CAMS reanalysis for the region of Poland. S-MESH estimates high $PM_{2.5}$ concentrations over large regions of Poland, whereas these high levels are not present in the CAMS forecast. This information from S-MESH aligns closely with the CAMS reanalysis. More such scenarios can be observed in Fig. 10 for different days. During these periods and generally during the winter months, residential heating contributes to a major extent towards $PM_{2.5}$ pollution over Poland and Eastern Europe (Hamer et al., 2023). With the CAMS forecast estimates as the base input, S-MESH can correct the forecast through the combination with other $PM_{2.5}$ indicators such as AOD, UAI, boundary layer height, weather conditions, and station observations. Even in the absence of AOD, S-MESH manages to capture these high concentration events over Poland and Eastern Europe (e.g., March 16, 2022 in Supplementary Fig. S4). Compared to the CAMS forecast, we can qualitatively observe improvements in spatial patterns over central and eastern Europe such as Poland, Romania, and others, where residential heating forms the major source of $PM_{2.5}$ pollution during the winter months. This is also typically underestimated in the CAMS reanalysis due to underestimation in the emission inventory (Tarrasón et al., 2022). From Figs. 9, 10a and 10b and Supplementary Fig. S4, more detailed spatial patterns in S-MESH can also be observed over these regions.

Agricultural emissions and traffic are the main contributors to $PM_{2.5}$ pollution levels in Western European regions such as the Netherlands, Belgium, Luxembourg, the United Kingdom, Denmark, and France. Fig. 10c shows high $PM_{2.5}$ levels during March 25, 2022 over these regions mainly arising from these sources (Hamer et al., 2023). Here again, the S-MESH spatial patterns match the reanalysis more closely, especially detecting the higher concentration levels, such as over Germany that are underestimated in the forecast.

Fig. 10d shows the $PM_{2.5}$ estimates for March 25, 2022 over Spain and Portugal from the three models, during an event of Saharan dust intrusion. In this case, the $PM_{2.5}$ levels shown by the forecast and S-MESH do not match that of the reanalysis. A wide band of $PM_{2.5}$ pollution with concentrations of $>40 \mu g/m^3$ is observed in Southern Spain in the reanalysis, while the forecast and S-MESH fail to capture the spatial extent of this pollution. S-MESH, which uses the CAMS forecast as its foundation, critically depends on the other input variables to

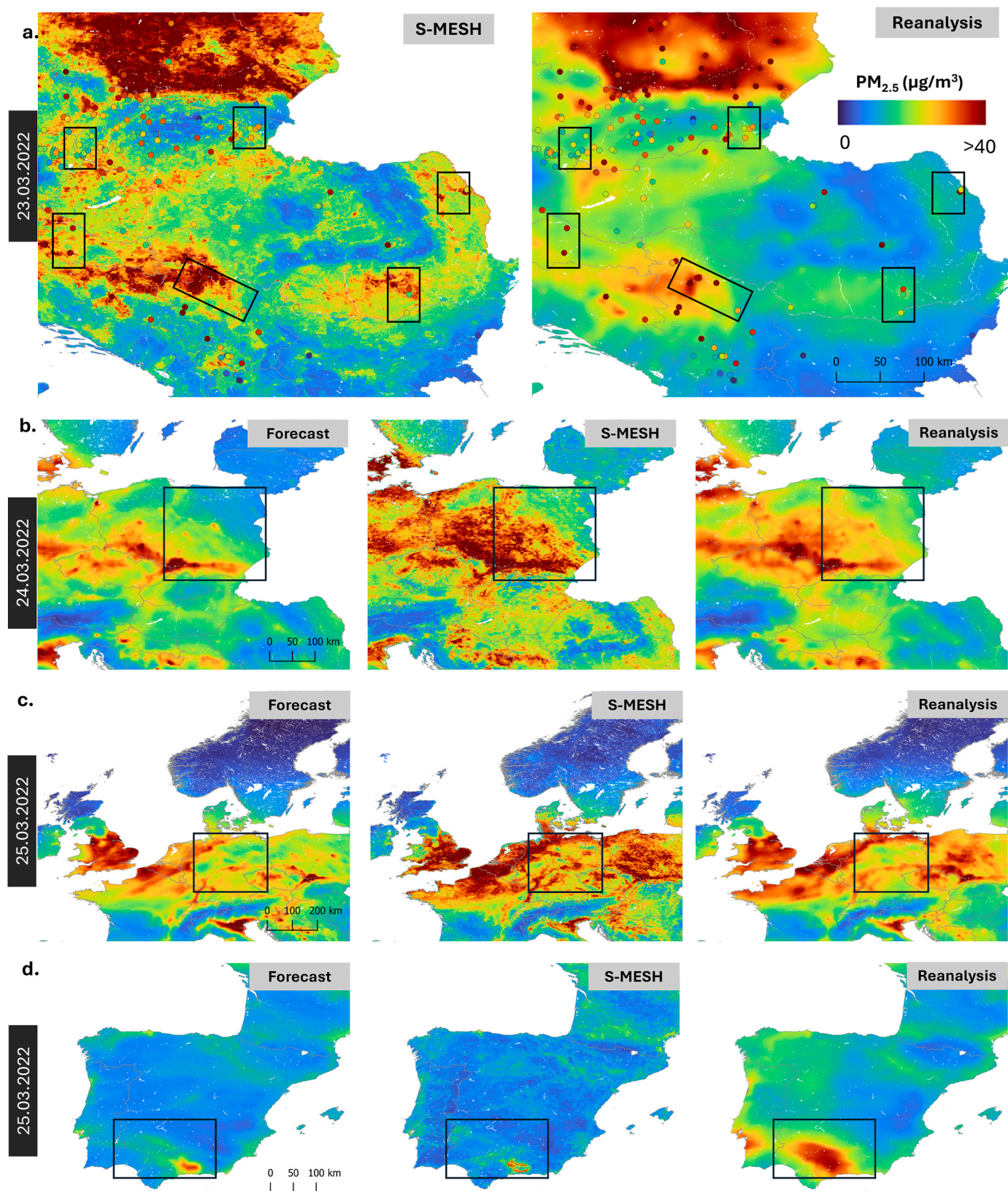


Fig. 10. Specific days of daily average $PM_{2.5}$ concentrations (in $\mu g/m^3$) from the CAMS forecast, S-MESH, and CAMS reanalysis during the March 2022 pollution episodes across different regions of the study area. Each row represents a different region or day, more specifically (a) Poland, Hungary, Romania, and southeastern Europe for March 23, 2022, (b) eastern Europe for March 24, 2022, (c) central and northern Europe for March 25, 2022, and (d) the Iberian Peninsula for March 25, 2022. The markers in panel (a) represent the monitoring station and their color indicates the measured concentration during the corresponding period. To better visualize the station measurements (only in Panel a), the forecast is omitted in panel a. The black rectangles highlight regions where differences can be seen.

capture such events and thus add value to the forecast. Dust events are normally captured by satellite AOD data or are sometimes indicated by the elevated values of UAI. However, AOD data are not always spatially continuous during the presence of clouds or thick dust layers. The absence of AOD data during this period over this region results in the S-MESH model being unable to capture this episode and the qualitative indicator of UAI is not sufficient for this purpose. Unlike the case of residential heating where the ML model acts as a bias corrector, this source of pollution cannot be captured by S-MESH if the event is not a part of either the forecast data or the satellite AOD. This highlights one of the disadvantages of the S-MESH for PM_{2.5} approach. A similar event over Spain is successfully captured by S-MESH for March 16, 2022 as it is also predicted in the forecast (Supplementary Figure S4). Including additional inputs that can potentially capture such events like information on dust AOD from Infrared Atmospheric Sounding Interferometer (IASI) or similar, synergized AOD from multiple satellite instruments that can provide data from more timestamps or using AOD from geostationary satellites, could help fill this gap in the model.

4. Conclusions

In this study, we describe and demonstrate our ML-based approach called S-MESH to estimate a daily hindcast of surface PM_{2.5} at a high spatial resolution of 1 km over Europe during the period 2021–2022. The method essentially downscales and corrects the 24-h CAMS regional PM_{2.5} forecast using additional input variables including AOD and AE from VIIRS, UAI from TROPOMI, meteorological conditions, population density, elevation, and temporal indicators. To obtain a spatially continuous estimate that is unaffected by gaps in the AOD data, a stacked ML approach based on the XGBoost algorithm is used. It successfully combines predictions from two separate models with and without AOD data and provides a spatially continuous estimate of surface PM_{2.5}. Comparative analysis of S-MESH's estimates with the CAMS regional forecast and the CAMS regional interim reanalysis is also performed to evaluate the potential of this approach.

Overall, S-MESH improves upon the CAMS forecast in terms of accuracy and spatial patterns and aligns well with the CAMS reanalysis while providing more spatial details. Annual estimates from S-MESH demonstrate higher accuracy than CAMS forecast and reanalysis. Further, the annual means from S-MESH, CAMS reanalysis, and station observations show higher PM_{2.5} levels over Poland and Eastern Europe compared to the forecast. In terms of daily estimates of S-MESH, their strengths are demonstrated particularly well over Poland, Hungary, and parts of Eastern Europe during high pollution levels, where PM_{2.5} primarily originates from residential heating. S-MESH also provides more detailed spatial variations of high pollution events over these regions. This could be observed well in the March 2022 pollution episode analyzed in the study where different sources contributed to high PM_{2.5} levels over Europe and were successfully captured by the S-MESH model. S-MESH also exhibits significant potential in capturing the day-to-day temporal variations of daily PM_{2.5} with better performance than reanalysis and can help identify periods of high pollution.

Using the CAMS forecast as one of the inputs, which also shows the highest variable importance in the model, ensures that physically and chemically resolved concentrations form the basis of the S-MESH estimates. However, this dependency also has some disadvantages in the current approach. S-MESH fails to capture certain PM_{2.5} events impacted by dust when the dust is neither predicted appropriately in the forecast nor available in the AOD data during the event. While UAI could also be an indicator of such events, it shows negligible impact in the model. S-MESH can be further improved for such scenarios to better match the reanalysis by using other inputs such as emission sources, for example, dust AOD from the Infrared Atmospheric Sounding Interferometer (IASI), and higher resolution AOD datasets, as well as AOD from geostationary satellites that can help capture certain events missed by single pass measurements of polar-orbiting satellites. Another area of focus for

S-MESH or similar works in the future but out of scope for the current study is analyzing the impact of unexpected events such as COVID-19 lockdowns on the current approach, particularly since the model is built on CAMS forecast.

S-MESH exhibits relatively low accuracy when estimating very low or background PM_{2.5} levels, particularly in regions like Finland where the pollution levels are typically very low. The CAMS reanalysis demonstrates higher accuracy over background regions, due to their defined capacity for capturing background pollution patterns. One potential way to improve the performance of S-MESH in such scenarios is by including more representative data in the training of the S-MESH model. This can be accomplished, for example by integrating low-cost sensor data, which is available at much higher spatial density than air quality monitoring stations, but this needs further research to assess its effectiveness. Another potential approach in improving S-MESH predictions at such levels would be using a weighted or other loss function such as quantile loss in XGBoost, that allows them to differently penalize the low prediction values. This, however, also needs further exploration to see how accuracy can be balanced across the concentration levels.

Despite some limitations, S-MESH can provide 1 km spatial resolution PM_{2.5} data for all of Europe and at an accuracy very close to - and in some cases surpassing - the CAMS reanalysis. Additionally, S-MESH uses station observations only for model training but not for prediction and as such enables a rapid analysis of the PM_{2.5} situation following pollution events. These features make S-MESH a valuable tool for near real-time analysis of pollution episodes and air quality assessments in the period when the reanalysis is not yet available. Its ability to map the spatial distribution and the temporal evolution of high PM_{2.5} levels makes it particularly useful for monitoring pollution patterns at a higher resolution. Thus, S-MESH can potentially further support air quality assessments, similar to how CAMS reanalysis is currently utilized. This approach can also be extended to other regional or global domains for surface PM_{2.5} estimations, where similar datasets are available.

CRedit authorship contribution statement

Shobitha Shetty: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Paul D. Hamer:** Writing – review & editing, Validation, Supervision, Resources, Investigation. **Kerstin Stebel:** Writing – review & editing, Validation, Supervision. **Arve Kylling:** Writing – review & editing, Validation, Supervision. **Amirhossein Hassani:** Writing – review & editing, Validation. **Terje Koren Berntsen:** Writing – review & editing, Supervision. **Philipp Schneider:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envres.2024.120363>.

Data availability

PM_{2.5} prediction maps generated from this study are available at <https://zenodo.org/records/13466510>. Other datasets used in this study are all publicly available.

References

- Apituley, A., Pedergrana, M., Sneep, M., Veeckind, J.P., Loyola, D., Zweers, D.S., 2018. In: Sentinel-5 Precursor/TROPOMI Level 2 Product User Manual UV Aerosol Index. The European Space Agency.
- Bertrand, J.-M., Meleux, F., Ung, A., Descombes, G., Colette, A., 2023. Technical note: improving the European air quality forecast of the Copernicus Atmosphere Monitoring Service using machine learning techniques. *Atmos. Chem. Phys.* 23 (9), 5317–5333. <https://doi.org/10.5194/acp-23-5317-2023>.
- Bi, J., Knowland, K.E., Keller, C.A., Liu, Y., 2022. Combining machine learning and numerical simulation for high-resolution PM_{2.5} concentration forecast. *Environ. Sci. Technol.* 56 (3), 1544–1556. <https://doi.org/10.1021/acs.est.1c05578>.
- CAMS, 2024a. CAMS European air quality forecasts. <https://ads.atmosphere.copernicus.eu/cdsapp#!/dataset/cams-europe-air-quality-forecasts?tab=overview>.
- CAMS, 2024b. CAMS European air quality reanalyses. <https://ads.atmosphere.copernicus.eu/cdsapp#!/dataset/cams-europe-air-quality-reanalyses?tab=overview>, 2024.
- CAMS, 2024c. CAMS policy support. <https://policy.atmosphere.copernicus.eu/reports/>.
- Casciaro, G., Cavaioia, M., Mazzino, A., 2022. Calibrating the CAMS European multi-model air quality forecasts for regional air pollution monitoring. *Atmos. Environ.* 287, 119259.
- Chen, T., Guestrin, C., 2016. XGBoost: A Scalable Tree Boosting System. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785–794. <https://doi.org/10.1145/2939672.2939785>.
- Chen, J., Qiu, S., Shang, J., Wilfrid, O.M.F., Liu, X., Tian, H., Boman, J., 2014. Impact of relative humidity and water soluble constituents of PM_{2.5} on visibility impairment in Beijing, China. *Aerosol Air Qual. Res.* 14 (1), 260–268.
- Chen, J., Yin, J., Zang, L., Zhang, T., Zhao, M., 2019. Stacking machine learning model for estimating hourly PM_{2.5} in China based on Himawari 8 aerosol optical depth data. *Sci. Total Environ.* 697, 134021. <https://doi.org/10.1016/j.scitotenv.2019.134021>.
- Cheng, Z., Luo, L., Wang, S., Wang, Y., Sharma, S., Shimadera, H., Wang, X., Bressi, M., de Miranda, R.M., Jiang, J., Zhou, W., Fajardo, O., Yan, N., Hao, J., 2016. Status and characteristics of ambient PM_{2.5} pollution in global megacities. *Environ. Int.* 89–90, 212–221. <https://doi.org/10.1016/j.envint.2016.02.003>.
- CIESIN, 2018. Gridded Population of the World, Version 4 (GPWv4): Population Density, Revision 11. NASA Socioeconomic Data and Applications Center (SEDAC). <https://doi.org/10.7927/H49C6VHW>.
- Climate Data Store, 2024. <https://cds.climate.copernicus.eu/#/home>.
- Coelho, S., Rafael, S., Lopes, D., Miranda, A.I., Ferreira, J., 2021. How changing climate may influence air pollution control strategies for 2030? *Sci. Total Environ.* 758, 143911. <https://doi.org/10.1016/j.scitotenv.2020.143911>.
- Copernicus, 2024. Copernicus data space Ecosystem. <https://dataspace.copernicus.eu/>.
- DAAC, L., 2023. LAADS DAAC - data archives. https://ladsweb.modaps.eosdis.nasa.gov/archive/allData/5200/AERDB_L2_VIIRS_NOAA20.
- Dhandapani, A., Iqbal, J., Kumar, R.N., 2023. Application of machine learning (individual vs stacking) models on MERRA-2 data to predict surface PM_{2.5} concentrations over India. *Chemosphere* 340. <https://doi.org/10.1016/j.chemosphere.2023.139966>.
- Didan, K., Barreto, A., 2018. VIIRS/NPP Vegetation Indices 16-Day L3 Global 500m SIN Grid V001. NASA EOSDIS Land Processes Distributed Active Archive Center. <https://doi.org/10.5067/VIIRS/VNP13A1.001> [Data set].
- Doxsey-Whitfield, E., MacManus, K., Adamo, S.B., Pistolesi, L., Squires, J., Borkovska, O., Baptista, S.R., 2015. Taking advantage of the improved availability of census data: a first look at the gridded population of the world, version 4. *Pap. Appl. Geogr.* 1 (3), 226–234. <https://doi.org/10.1080/23754931.2015.1014272>.
- ECMWF, 2024. CAMS Regional: European air quality reanalyses data documentation. <https://confluence.ecmwf.int/display/CKB/CAMS+Regional%3A+European+air+quality+reanalyses+data+documentation#heading-Howtoacknowledgeciteandefertothedata>.
- EEA, 2020. Air quality in Europe - 2020 report. <https://doi.org/10.2800/786656>.
- EEA, 2021. Air quality in Europe 2021. <https://www.eea.europa.eu/publications/air-quality-in-europe-2022>.
- EEA, 2022. Air quality in Europe 2022. <https://doi.org/10.2800/488115>.
- EEA, 2023a. Air quality e-reporting - European Environment Agency. <https://www.eea.europa.eu/en/datahub/datahubitem-view/3b390c9c-f321-490a-b25a-ae93b2ed80c1>.
- EEA, 2023b. Europe's air quality status 2023. <https://doi.org/10.2800/59526>.
- EEA, 2024. Europe's air quality status 2024. <https://doi.org/10.2800/5970>.
- European Union, 2008. Directive 2008/50/EC of the European Parliament and of the Council of 21 May 2008 on ambient air quality and cleaner air for Europe. In: *Official Journal of the European Union* (Issue L 152). <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32008L0050>.
- European Union, 2022. Proposal for a revision of the ambient air quality directives. In: European Parliamentary Research Service, vol. 347. Issue October. [https://www.europarl.europa.eu/RegData/etudes/BRIE/2022/734679/EPRS_BRI\(2022\)734679_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/BRIE/2022/734679/EPRS_BRI(2022)734679_EN.pdf).
- Favez, O., Cachier, H., Sciare, J., Sarda-Estève, R., Martinon, L., 2009. Evidence for a significant contribution of wood burning aerosols to PM_{2.5} during the winter season in Paris, France. *Atmos. Environ.* 43 (22), 3640–3644. <https://doi.org/10.1016/j.atmosenv.2009.04.035>.
- Geiss, A., Silva, S.J., Hardin, J.C., 2022. Downscaling atmospheric chemistry simulations with physically consistent deep learning. *Geosci. Model Dev. (GMD)* 15 (17), 6677–6694. <https://doi.org/10.5194/gmd-15-6677-2022>.
- Geng, G., Zhang, Q., Martin, R.V., van Donkelaar, A., Huo, H., Che, H., Lin, J., He, K., 2015. Estimating long-term PM_{2.5} concentrations in China using satellite-based aerosol optical depth and a chemical transport model. *Rem. Sens. Environ.* 166, 262–270. <https://doi.org/10.1016/j.rse.2015.05.016>.
- Goldberg, D.L., Gupta, P., Wang, K., Jena, C., Zhang, Y., Lu, Z., Streets, D.G., 2019. Using gap-filled MAIAC AOD and WRF-Chem to estimate daily PM_{2.5} concentrations at 1 km resolution in the Eastern United States. *Atmos. Environ.* 199, 443–452. <https://doi.org/10.1016/j.atmosenv.2018.11.049>.
- Guan, X., Wang, M., Du, T., Tian, P., Zhang, N., Shi, J., Chang, Y., Zhang, L., Zhang, M., Song, X., Sun, Y., 2021. Wintertime aerosol optical properties in Lanzhou, Northwest China: emphasis on the rapid increase of aerosol absorption under high particulate pollution. *Atmos. Environ.* 246, 118081. <https://doi.org/10.1016/j.atmosenv.2020.118081>.
- Gupta, P., Christopher, S.A., 2009. Particulate matter air quality assessment using integrated surface, satellite, and meteorological products: multiple regression approach. *J. Geophys. Res. Atmos.* 114 (D14), 14205. <https://doi.org/10.1029/2008JD011496>.
- Gupta, P., Zhan, S., Mishra, V., Aekakkararungroj, A., Markert, A., Paibong, S., Chishtie, F., 2021. Machine learning algorithm for estimating surface pm2.5 in Thailand. *Aerosol Air Qual. Res.* 21 (11). <https://doi.org/10.4209/AAQR.210105>.
- Hamer, P., Fjærraa, A.M., Tarrasón, L., Soares, J., Meleux, F., Colette, A., Ung, A., Raux, B., Kuenen, J., 2023. Copernicus atmosphere monitoring Service interim annual assessment report on European air quality in 2022. https://policy.atmosphere.copernicus.eu/reports/pdf/CAMS271_2021SCX_D1.1.1_202306_2022_Interim_Assessment_Report_v3.pdf.
- Hersbach, H., Bell, B., Berrisford, P., Biavati, G., Horányi, A., Muñoz Sabater, J., Nicolas, J., Peubey, C., Radu, R., Rozum, I., Schepers, D., Simmons, A., Soci, C., Dee, D., Thépaut, J.-N., 2023. ERA5 hourly data on pressure levels from 1940 to present. Copernicus Climate Change Service (C3S) Climate Data Store (CDS). <https://doi.org/10.24381/cds.bd0915c6>.
- Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., Nicolas, J., Peubey, C., Radu, R., Schepers, D., Simmons, A., Soci, C., Abdalla, S., Abellan, X., Balsamo, G., Bechtold, P., Biavati, G., Bidlot, J., Bonavita, M., et al., 2020. The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.* 146 (730), 1999–2049. <https://doi.org/10.1002/qj.3803>.
- Hsu, C., 2018. Heritage and summary of VIIRS Deep Blue and SOAR aerosol property retrieval algorithms. <https://doi.org/10.1002/2017JD027412>.
- Hsu, N.C., Lee, J., Sayer, A.M., Kim, W., Bettenhausen, C., Tsay, S.-C., 2019. VIIRS deep blue aerosol products over land: extending the EOS long-term aerosol data records. *J. Geophys. Res. Atmos.* 124 (7), 4026–4053. <https://doi.org/10.1029/2018JD029688>.
- Hu, X., Belle, J.H., Meng, X., Wildani, A., Waller, L.A., Strickland, M.J., Liu, Y., 2017. Estimating PM_{2.5} concentrations in the conterminous United States using the random forest approach. *Environ. Sci. Technol.* 51 (12), 6936–6944. <https://doi.org/10.1021/acs.est.7b01210>.
- Hu, X., Waller, L.A., Al-Hamdan, M.Z., Crosson, W.L., Estes, M.G., Estes, S.M., Quattrocchi, D.A., Sarnat, J.A., Liu, Y., 2013. Estimating ground-level PM_{2.5} concentrations in the southeastern U.S. using geographically weighted regression. *Environ. Res.* 121, 1–10. <https://doi.org/10.1016/j.envres.2012.11.003>.
- Ibrahim, S., Landa, M., Pešek, O., Brodský, L., Halounová, L., 2022. Machine learning-based approach using open data to estimate PM_{2.5} over Europe. *Rem. Sens.* 14 (14), 1–13. <https://doi.org/10.3390/rs14143392>.
- Jiang, T., Chen, B., Nie, Z., Ren, Z., Xu, B., Tang, S., 2021. Estimation of hourly full-coverage PM_{2.5} concentrations at 1-km resolution in China using a two-stage random forest model. *Atmos. Res.* 248 (February 2020), 105146. <https://doi.org/10.1016/j.atmosres.2020.105146>.
- Jin, H., Chen, X., Zhong, R., Liu, M., 2022. Influence and prediction of PM_{2.5} through multiple environmental variables in China. *Sci. Total Environ.* 849 (June), 157910. <https://doi.org/10.1016/j.scitotenv.2022.157910>.
- Jin, J., Henzing, B., Segers, A., 2023. How aerosol size matters in aerosol optical depth (AOD) assimilation and the optimization using the Ångström exponent. *Atmos. Chem. Phys.* 23 (2), 1641–1660.
- Kaffashzadeh, N., Aliakbari Bidokhti, A.A., 2024. Assessment of surface ozone products from downscaled CAMS reanalysis and CAMS daily forecast using urban air quality monitoring stations in Iran. *Geosci. Model Dev. (GMD)* 17 (10), 4155–4179. <https://doi.org/10.5194/gmd-17-4155-2024>.
- Kuenen, J., Dellaert, S., Visschedijk, A., Jalkanen, J.-P., Super, I., Denier van der Gon, H., 2022. CAMS-REG-v4: a state-of-the-art high-resolution European emission inventory for air quality modelling. *Earth Syst. Sci. Data* 14 (2), 491–515.

- Liang, C.-K., West, J.J., Silva, R.A., Bian, H., Chin, M., Davila, Y., Dentener, F.J., Emmons, L., Flemming, J., Folberth, G., 2018. HTAP2 multi-model estimates of premature human mortality due to intercontinental transport of air pollution and emission sectors. *Atmos. Chem. Phys.* 18 (14), 10497–10520.
- Liu, Y., Park, R.J., Jacob, D.J., Li, Q., Kilaru, V., Sarnat, J.A., 2004. Mapping annual mean ground-level PM_{2.5} concentrations using Multiangle Imaging Spectroradiometer aerosol optical thickness over the contiguous United States. *J. Geophys. Res. Atmos.* 109 (D22).
- Lundberg, S.M., Erion, G., Chen, H., DeGrave, A., Prutkin, J.M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N., Lee, S.I., 2020. From local explanations to global understanding with explainable AI for trees. *Nat. Mach. Intell.* 2 (1), 56–67. <https://doi.org/10.1038/s42256-019-0138-9>.
- Lundberg, S.M., Lee, S.-I., 2017. A unified approach to interpreting model predictions. In: Guyon, I., Von Luxburg, U., Bengio, S., Wallach, H., Fergus, R., Vishwanathan, S., Garnett, R. (Eds.), *Advances in Neural Information Processing Systems*, vol. 30. Curran Associates, Inc. In: <https://proceedings.neurips.cc/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf>.
- Manisalidis, I., Stavropoulou, E., Stavropoulos, A., Bezirtzoglou, E., 2020. Environmental and health impacts of air pollution: a review. In: *Frontiers in Public Health*, vol. 8. Frontiers Media S.A., p. 14. <https://doi.org/10.3389/fpubh.2020.00014>.
- Maréchal, V., Peuch, V.H., Andersson, C., Andersson, S., Arteta, J., Beekmann, M., Benedictow, A., Bergström, R., Bessagnet, B., Cansado, A., Chéroux, F., Colette, A., Coman, A., Curier, R.L., Van Der Gon, H.A.C.D., Drouin, A., Elbern, H., Emili, E., Engelen, R.J., et al., 2015. A regional air quality forecasting system over Europe: the MACC-II daily ensemble production. *Geosci. Model Dev. (GMD)* 8 (9), 2777–2813. <https://doi.org/10.5194/gmd-8-2777-2015>.
- McDuffie, E.E., Smith, S.J., O'Rourke, P., Tibrewal, K., Venkataraman, C., Marais, E.A., Zheng, B., Crippa, M., Brauer, M., Martin, R.V., 2020. A global anthropogenic emission inventory of atmospheric pollutants from sector- and fuel-specific sources (1970–2017): an application of the Community Emissions Data System (CEDS). *Earth Syst. Sci. Data* 12 (4), 3413–3442. <https://doi.org/10.5194/essd-12-3413-2020>.
- Miranda, A., Silveira, C., Ferreira, J., Monteiro, A., Lopes, D., Relvas, H., Borrego, C., Roebeling, P., 2015. Current air quality plans in Europe designed to support air quality management policies. *Atmos. Pollut. Res.* 6 (3), 434–443. <https://doi.org/10.5094/APR.2015.048>.
- Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G., Balsamo, G., Boussetta, S., Choulga, M., Harrigan, S., Hersbach, H., Martens, B., Miralles, D.G., Piles, M., Rodríguez-Fernández, N.J., Zsoter, E., Buontempo, C., Thépaut, J.-N., 2021. ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. *Earth Syst. Sci. Data* 13 (9), 4349–4383. <https://doi.org/10.5194/essd-13-4349-2021>.
- NASA, 2022. The Deep Blue aerosol project: aerosol retrievals from S-NPP VIIRS. https://ladsweb.modaps.eosdis.nasa.gov/api/v2/content/archives/DocumentArchive/ScienceDataProductDocumentation/VIIRS_Deep_Blue_Aerosol_User_Guide.v1.1.pdf.
- Peuch, V.-H., Engelen, R., Rixen, M., Dee, D., Flemming, J., Suttie, M., Ades, M., Agustí-Panareda, A., Ananasso, C., Andersson, E., Armstrong, D., Barré, J., Bousserez, N., Dominguez, J.J., Garrigues, S., Inness, A., Jones, L., Kipling, Z., Letertre-Danczak, J., et al., 2022. The Copernicus atmosphere monitoring Service: from research to operations. *Bull. Am. Meteorol. Soc.* 103 (12), E2650–E2668. <https://doi.org/10.1175/BAMS-D-21-0314.1>.
- Pope III, C.A., Lefler, J.S., Ezzati, M., Higbee, J.D., Marshall, J.D., Kim, S.-Y., Bechle, M., Gilliat, K.S., Vernon, S.E., Robinson, A.L., 2019. Mortality risk and fine particulate air pollution in a large, representative cohort of US adults. *Environ. Health Perspect.* 127 (7), 77007.
- Salah Eddine, S., Drissi, L.B., Mejjad, N., Mabrouki, J., Romanov, A.A., 2024. Machine learning models application for spatiotemporal patterns of particulate matter prediction and forecasting over Morocco in north of Africa. *Atmos. Pollut. Res.* 15 (9), 102239. <https://doi.org/10.1016/j.apr.2024.102239>.
- Sayer, A.M., Hsu, N.C., Lee, J., Kim, W.V., Dutcher, S.T., 2019. Validation, stability, and consistency of MODIS collection 6.1 and VIIRS version 1 deep blue aerosol data over land. *J. Geophys. Res. Atmos.* 124 (8), 4658–4688. <https://doi.org/10.1029/2018JD029598>.
- Shao, P., Xin, J., An, J., Kong, L., Wang, B., Wang, J., Wang, Y., Wu, D., 2017. The empirical relationship between PM_{2.5} and AOD in Nanjing of the Yangtze river delta. *Atmos. Pollut. Res.* 8 (2), 233–243.
- Shao, Y., Ma, Z., Wang, J., Bi, J., 2020. Estimating daily ground-level PM_{2.5} in China with random-forest-based spatiotemporal kriging. *Sci. Total Environ.* 740, 139761. <https://doi.org/10.1016/j.scitotenv.2020.139761>.
- Shetty, S., Schneider, P., Stebel, K., David Hamer, P., Kylling, A., Koren Bernsten, T., 2024. Estimating surface NO₂ concentrations over Europe using Sentinel-5P TROPOMI observations and machine learning. *Rem. Sens. Environ.* 312, 114321. <https://doi.org/10.1016/j.rse.2024.114321>.
- Sluijterman, L., Kreuwel, F., Cator, E., Heskes, T., 2024. Composite Quantile Regression With XGBoost Using the Novel ArcTan Pinball Loss. *ArXiv Preprint ArXiv: 2406.02293*, 24. <https://doi.org/10.48550/arXiv.2406.02293>.
- Sokhi, R.S., Moussiopoulos, N., Baklanov, A., Bartzis, J., Coll, I., Finardi, S., Friedrich, R., Geels, C., Grönholm, T., Halenka, T., Ketzel, M., Maragkidou, A., Matthias, V., Moldanova, J., Ntziachristos, L., Schäfer, K., Suppan, P., Tsegas, G., Carmichael, G., et al., 2022. Advances in air quality research – current and emerging challenges. *Atmos. Chem. Phys.* 22 (7), 4615–4703. <https://doi.org/10.5194/ACP-22-4615-2022>.
- Song, W., Jia, H., Huang, J., Zhang, Y., 2014. A satellite-based geographically weighted regression model for regional PM_{2.5} estimation over the Pearl River Delta region in China. *Rem. Sens. Environ.* 154, 1–7.
- Tang, D., Zhan, Y., Yang, F., 2024. A review of machine learning for modeling air quality: overlooked but important issues. *Atmos. Res.* 300 (January), 107261. <https://doi.org/10.1016/j.atmosres.2024.107261>.
- Tarrasón, L., Hamer, P.D., Mari Fjæraa, Ann, Meleux, F., Colette, A., Ung, A., Kuenen, J., Arjan, D., Guevara, M., 2022. Copernicus atmosphere monitoring Service interim annual assessment report for 2021. https://policy.atmosphere.copernicus.eu/reports/CAMS271_IAR_2021_Final.pdf.
- Unik, M., Sitanggang, I.S., Syaifulna, L., Jaya, I.N.S., 2023. PM_{2.5} estimation using machine learning models and satellite data: a literature review. *Int. J. Adv. Comput. Sci. Appl.* 14 (5), 359–370. <https://doi.org/10.14569/IJACSA.2023.0140538>.
- Van Donkelaar, A., Martin, R.V., Brauer, M., Hsu, N.C., Kahn, R.A., Levy, R.C., Lyapustin, A., Sayer, A.M., Winker, D.M., 2016. Global estimates of fine particulate matter using a combined geophysical-statistical method with information from satellites, models, and monitors. *Environ. Sci. Technol.* 50 (7), 3762–3772. <https://doi.org/10.1021/acs.est.5b05833>.
- van Donkelaar, A., Martin, R.V., Brauer, M., Kahn, R., Levy, R., Verduzco, C., Villeneuve, P.J., 2010. Global estimates of ambient fine particulate matter concentrations from satellite-based aerosol optical depth: development and application. *Environ. Health Perspect.* 118 (6), 847–855. <https://doi.org/10.1289/ehp.0901623>.
- Varga-Balogh, A., Leelőssy, Á., Lagzi, I., Mészáros, R., 2020. Time-dependent downscaling of PM_{2.5} predictions from CAMS air quality models to urban monitoring sites in Budapest. *Atmosphere* 11 (6), 1–15. <https://doi.org/10.3390/atmos11060669>.
- Wang, J., Christopher, S.A., 2003. Intercomparison between satellite-derived aerosol optical thickness and PM_{2.5} mass: implications for air quality studies. *Geophys. Res. Lett.* 30 (21).
- Wei, J., Li, Z., Cribb, M., Huang, W., Xue, W., Sun, L., Guo, J., Peng, Y., Li, J., Lyapustin, A., Liu, L., Wu, H., Song, Y., 2020. Improved 1km resolution PM_{2.5} estimates across China using enhanced space-time extremely randomized trees. *Atmos. Chem. Phys.* 20 (6), 3273–3289. <https://doi.org/10.5194/acp-20-3273-2020>.
- Wei, J., Li, Z., Lyapustin, A., Sun, L., Peng, Y., Xue, W., Su, T., Cribb, M., 2021. Reconstructing 1-km-resolution high-quality PM_{2.5} data records from 2000 to 2018 in China: spatiotemporal variations and policy implications. *Rem. Sens. Environ.* 252 (January 2020), 112136. <https://doi.org/10.1016/j.rse.2020.112136>.
- Wu, W., Zhang, M., Ding, Y., 2020. Exploring the effect of economic and environment factors on PM_{2.5} concentration: a case study of the Beijing-Tianjin-Hebei region. *J. Environ. Manag.* 268, 110703.
- Xiao, Q., Chang, H.H., Geng, G., Liu, Y., 2018. An ensemble machine-learning model to predict historical PM_{2.5} concentrations in China from satellite data. *Environ. Sci. Technol.* 52 (22), 13260–13269. <https://doi.org/10.1021/acs.est.8b02917>.
- Xiao, Y., Wang, Y., Yuan, Q., He, J., Zhang, L., 2022. Generating a long-term (2003–2020) hourly 0.25° global PM_{2.5} dataset via spatiotemporal downscaling of CAMS with deep learning (DeepCAMS). *Sci. Total Environ.* 848 (July), 157747. <https://doi.org/10.1016/j.scitotenv.2022.157747>.
- Xu, G., Ren, X., Xiong, K., Li, L., Bi, X., Wu, Q., 2020. Analysis of the driving factors of PM_{2.5} concentration in the air: a case study of the Yangtze River Delta, China. *Ecol. Indic.* 110 (July 2019), 105889. <https://doi.org/10.1016/j.ecolind.2019.105889>.
- Xu, X., Zhang, C., Liang, Y., 2021. Review of satellite-driven statistical models PM_{2.5} concentration estimation with comprehensive information. *Atmos. Environ.* 256 (June 2020), 118302. <https://doi.org/10.1016/j.atmosenv.2021.118302>.
- Yamazaki, D., Ikeshima, D., Tawatari, R., Yamaguchi, T., O'Loughlin, F., Neal, J.C., Sampson, C.C., Kanea, S., Bates, P.D., 2017. A high-accuracy map of global terrain elevations. *Geophys. Res. Lett.* 44 (11), 5844–5853. <https://doi.org/10.1002/2017GL072874>.
- Yang, N., Shi, H., Tang, H., Yang, X., 2022. Geographical and temporal encoding for improving the estimation of PM_{2.5} concentrations in China using end-to-end gradient boosting. *Rem. Sens. Environ.* 269, 112828. <https://doi.org/10.1016/j.rse.2021.112828>.
- Yang, S., Fang, D., Chen, B., 2019. Human health impact and economic effect for PM_{2.5} exposure in typical cities. *Appl. Energy* 249, 316–325.
- Zang, Z., Li, D., Guo, Y., Shi, W., Yan, X., 2021. Superior PM_{2.5} estimation by integrating aerosol fine mode data from the Himawari-8 satellite in deep and classical machine learning models. In: *Remote Sensing*, vol. 13. <https://doi.org/10.3390/rs13142779>, 14.
- Zhang, H., Kondragunta, S., 2021. Daily and hourly surface PM_{2.5} estimation from satellite AOD. *Earth Space Sci.* 8 (3). <https://doi.org/10.1029/2020EA001599>.
- Zhang, H., Wang, J., García, L.C., Zhou, M., Ge, C., Plessel, T., Szykman, J., Levy, R.C., Murphy, B., Spero, T.L., 2022. Improving surface PM_{2.5} forecasts in the United States using an ensemble of chemical transport model outputs: 2. Bias correction with satellite data for rural areas. *J. Geophys. Res. Atmos.* 127 (1), e2021JD035563.
- Zhang, T., Zhu, Z., Gong, W., Zhu, Z., Sun, K., Wang, L., Huang, Y., Mao, F., Shen, H., Li, Z., Xu, K., 2018. Estimation of ultrahigh resolution PM_{2.5} concentrations in urban areas using 160 m Gaofen-1 AOD retrievals. *Rem. Sens. Environ.* 216 (July), 91–104. <https://doi.org/10.1016/j.rse.2018.06.030>.
- Zhang, Y., Li, Z., Bai, K., Wei, Y., Xie, Y., Zhang, Y., Ou, Y., Cohen, J., Zhang, Y., Peng, Z., Zhang, X., Chen, C., Hong, J., Xu, H., Guang, J., Lv, Y., Li, K., Li, D., 2021. Satellite remote sensing of atmospheric particulate matter mass concentration: advances, challenges, and perspectives. *Fundam. Res.* 1 (3), 240–258. <https://doi.org/10.1016/j.fmr.2021.04.007>.